Upgrading the Global Financial Safety Net: Implications of a Synthetic Reserve Deposit*

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Abstract

Sovereigns self-insure via reserve accumulation, prioritizing liquidity over yield. This practice imposes opportunity costs and can amplify the convenience premium on traditional reserve assets. This paper evaluates Synthetic Reserve Deposits (SRDs), a pooling mechanism designed to improve the return-liquidity trade-off. Participants would hold floating-value claims on a vehicle comprising a settlement buffer, an investment portfolio, and, in some designs, an optional collateralized liquidity window. The architecture separates the allocation of portfolio risk from the provision of short-term foreign-currency liquidity: market risk remains with SRD holders, while liquidity is provided through redemption at contemporaneous executable value or through secured advances. Simulations illustrate that the structure can shift the return-liquidity frontier outward under explicit downside-risk constraints. General equilibrium modeling suggests that, if implemented at sufficient scale and under appropriate conditions, reallocating reserve demand through this mechanism could increase the effective supply of reserve claims and affect the global financial cycle, creating aggregate welfare gains and possibly Pareto improvements depending on initial global asset positions.

JEL Classification F31, F33, F41, G11

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Introduction

The modern International Monetary and Financial System features a scarcity of safe assets. Countries facing external risk value foreign exchange reserves that are stable in value and immediately deployable. Correspondingly, they self-insure through large holdings of short-duration reserve assets. However, since the supply of assets that are simultaneously safe, liquid, and yield a reasonable return is limited, decentralized reserve accumulation increases the opportunity cost of holding reserves and may also generate global distortions (Caballero, Farhi, and Gourinchas, 2017, Das et al., 2022). Can we do better?

This paper studies a class of arrangements through which countries could lower the cost of meeting reserve liquidity and safety objectives by pooling part of their foreign-exchange reserves. These arrangements, which we label "Synthetic Reserve Deposits" (SRDs), aim to maximize returns subject to the safety and liquidity requirements imposed by the definition of a reserve asset. We show that such a synthesizing exercise can find asset portfolios which simultaneously (i) meet the definition of a reserve asset, (ii) are not fully comprised of reserve assets and (iii) obtain higher returns than individual reserve assets, without necessarily taking on more risk. By enabling coordination of reserve accumulation and management decisions, these arrangements can expand the effective supply of reserve-eligible assets, creating aggregate gains.

The key distinguishing feature of the arrangements we consider is that diversification and contingent liquidity management are shifted from the balance sheet of the individual reserve holder to that of a pooled vehicle. Like investors in a mutual fund, participant countries would hold a floating-value claim issued by the pool, while the pool combines a cash buffer for settlement and routine redemptions with a diversified investment portfolio. The scheme is then able to provide gains along three dimensions: improving on the risk-return frontier, reducing asset turnover and transaction costs, and potentially also enabling short-term liquidity provision against long-term asset values.

First, an arrangement managed by an official third-party entity can combine assets in a manner that retains market confidence about the portfolio remaining a credible reserve asset overall. Maintaining the reserve-asset nature of foreign-exchange reserves is a central challenge in portfolio management by central banks. To unquestionably communicate the reserve-asset nature of their reserves, central banks tend to invest the predominant share of their reserves into assets that are individually reserve assets. An advantage of this strategy is that active trading and valuation changes impacting portfolio composition do not endanger the reserve-asset qualification of the portfolio, since each security held is reserves. However, it impedes portfolio constructions which are not entirely made of securities that are themselves reserve assets. Such portfolios are much more challenging to communicate, as any trade or change in value would shift underlying portfolio weights and put the reserve-asset nature of the portfolio in question.[1] By enforcing these rules on itself, an arrangement like the SRD can select investments to maximize returns subject to the constraint that the resulting portfolio—rather than the individual securities—satisfies the safety and liquidity characteristics of a reserve asset. Moreover, careful portfolio selection would take into account participants' stochastic discount factors: risk is not only undesirable per se, but it is also especially hurtful if liquidity needs are more acute at the times at which the portfolio has fallen in value (the possibility of fire sales, of course, worsens this issue). The SRD design we consider seeks a portfolio that increases returns relative to current reserve assets while retaining reserve-asset qualification and the "good friend" property—that is, providing payoffs or liquidity services that are especially valuable in adverse states—in the sense of Brunnermeier, Merkel, and Sannikov (2024).

Second, a pooled arrangement can reduce transactions costs. By centralizing trading, custody, and collateral management arrangements, the SRD can economize on fixed and variable costs that reserve holders would otherwise bear separately. To the extent that participant's liquidity needs are not fully synchronized, the pooled structure can net flows internally and reduce the frequency and size of external transactions with private intermediaries. Consequently, fewer trades would be needed which would limit transaction costs. While this point applies under tranquil market conditions, any friction on the side of private banks and intermediaries, or stemming from trade timing (such as fire sales or times of market turbulence), would magnify this source of gains.

A third and related layer could enhance the liquidity function through additional facilities providing liquidity secured by a country's SRD holdings. These would work like a discount window, allowing participants to borrow short-term against SRD units. Conservative margining and haircuts would guarantee that such borrowing is kept at minimal risk to the scheme. But by allowing countries to raise cash against the underlying investment portfolio rather than through outright asset sales, the facility would reduce the need to liquidate the savings portfolio during periods of temporary asset valuation slumps. Consequently, it would tide reserve holders through asset price movements which are truly temporary such as distressed market conditions. Any losses on the underlying portfolio would remain with SRD holders, further minimizing risks for the facility itself.

The purpose of this paper is analytical rather than prescriptive. We do not propose a fully specified operational facility, nor do we treat the SRD as a device for unconstrained yield enhancement. We employ historical backtests and stochastic simulations as numerical illustrations of how such a mechanism could be designed and calibrated.

[1] End-of-quarter effects are well known in banking regulation, making it difficult for a well-funded regulator to enforce portfolio-composition rules at a high frequency, which is critical for foreign reserves. On the other hand, after publication of the scheme's portfolio weights, further gains could be enabled if central banks could likewise credibly communicate holdings in accordance to these weights, perhaps with private third-party monitoring.

The SRD is designed to retain liquidity and safety, properties of reserve assets. While it is therefore expected that the SRD will qualify as a reserve asset, a full assessment of its reserve status will depend on implementation details that go beyond the scope of the present paper.[2]

The quantitative illustrations are conducted in U.S. dollar-denominated assets. This choice reflects the central role of dollar liquidity in many stress episodes and keeps the reserve-liquidity question separate from the currency-allocation problem.[3] We base our analysis on a scheme remaining small relative to the market (about USD 50bn in AUM), with a portfolio that is mostly made up of long-dated A sovereign and sub-sovereign bonds, long-dated A corporate bonds, along with an equity tranche (about 16%) and a liquidity tranche (short-dated Treasuries, about 10%), as detailed in Table 1.[4]

The exercises suggest three broad conclusions. First, the scheme with internalized investment management and contingent liquidity can improve the return profile of reserve holdings relative to a strategy concentrated in conventional reserve assets. Second, the illustrative SRD portfolio has return properties closer to those of reserve assets than to those of unconstrained risky assets. In this sense, the SRD reproduces part of the state-contingent services associated with safe assets in Brunnermeier, Merkel, and Sannikov (2024), while allowing the underlying pool to be more diversified than a conventional reserve portfolio. Countries holding a floating-value claim on an SRD portfolio composed of the above long-dated bonds and equities will make losses after jumps in interest rates on the bond categories and falls in equity prices, but the portfolio weights help reduce the size of such losses. Third, when liquidity is supplied against collateral rather than through forced sales, short-horizon tail risk can remain limited under conservative haircut and margining rules, so temporary valuation losses need not disrupt liquidity access. Ultimately, our portfolio selection exercise establishes that annual excess returns of the order of 250bps can be achieved while retaining the safety and liquidation properties of traditional reserve assets.

The scale of this channel is important. The simulations and design considered in this paper are consistent with an SRD that does not significantly move global asset prices. At small scale, the main benefits of an SRD are likely to be participant-level. However, a large scheme would have global macroeconomic implications. By increasing the effective supply of reserve-eligible

[2] For an SRD claim to be usable as a reserve asset in practice, its legal form and operating rules would have to ensure that the claim is an external foreign-currency asset controlled by the monetary authority and redeemable or liquidatable at contemporaneous market value within the time frame required for ready availability. Technological developments could potentially mitigate time-frame considerations, for example tokenized asset holdings could ensure quasi instantaneous liquidation at least into tokenized deposits.

[3] Nothing in the mechanism is specific to the dollar, however: the same balance-sheet logic could be applied to multi-currency portfolios or to a portfolio benchmarked to the SDR basket, with currency composition treated as an additional design choice.

[4] Such an analysis naturally assumes that the asset return distributions and correlations of different kinds of assets remain stable over time.

assets, the scheme could reduce the pressure that decentralized reserve accumulation places on the stock of traditional safe assets.

General equilibrium modeling of the global economy in a companion paper establishes that reallocating reserve demand through this mechanism can indeed increase the effective supply of reserve claims. Moreover, by doing so, it can affect the global financial cycle by altering global firms' decisions about their borrowing in different currencies. This would create welfare gains in the aggregate and could yield Pareto improvements depending on the initial global asset positions. Specifically, Pareto improvements would arise in configurations where the U.S. owns comparatively a greater share of global firms than of global financial intermediaries.

The remainder of the paper proceeds as follows. Section 2 situates the paper in the related literature on safe-asset scarcity, reserve management, and decentralized self-insurance. Section 3 documents the economics of reserve self-insurance by reviewing the insurance value of reserves, the fragmented structure of the Global Financial Safety Net, and the quasi-fiscal cost of precautionary reserve accumulation. Section 4 sets out the balance-sheet mechanics of the SRD and the role of the optional collateralized liquidity window. Section 5 presents quantitative illustrations of the benchmark portfolio design and its risk-return properties. Section 6 studies liquidity frictions, haircut calibration, margining, and short-horizon tail risk. Section 7 examines aggregate scale, country coverage, and potential market impact. Section 8 discusses broader general-equilibrium implications and the conditions under which an SRD arrangement can improve outcomes in the global economy. The appendices provide details on the data, simulation design, stochastic discount factor construction, and efficient-frontier analysis.

2. Related Literature

This paper speaks to three related literatures: safe-asset scarcity and the international monetary system, the asset-pricing foundations of safe assets, and the inefficiencies associated with decentralized reserve accumulation.

First, the analysis builds on work on the structural shortage of safe assets and the architecture of the international monetary system. Caballero, Farhi, and Gourinchas (2017) document a persistent imbalance between the global demand for safe stores of value and the capacity of advanced economies to supply them. In their account, the demand for stores of values arising from EMDEs is an important source of that pressure. The same mechanism depresses equilibrium safe rates and raises the quasi-fiscal cost of reserve accumulation. Related policy work emphasizes that the Global Financial Safety Net (GFSN) remains fragmented, leaving many developing economies reliant on unilateral reserve accumulation despite the existence of other formal layers (International Monetary Fund, 2023; Acalin et al., 2026). In the 2023 Mundell-Fleming lecture, one of us argued that this fragmentation creates substantial inefficiencies and highlights the role of multilateral arrangements that can provide predictable contingent liquidity, including swap lines and international repo facilities (Gourinchas, 2023). This paper contributes to that literature by specifying a balance-sheet mechanism through which pooling and collateralized liquidity provision can increase the effective supply of reserve claims without requiring new issuance of underlying sovereign debt.

Second, the design of the SRD draws on the asset-pricing microfoundations of safe assets. Brunnermeier, Merkel, and Sannikov (2024) emphasize that safe assets provide state-contingent liquidity services: they are especially valuable when the holder is hit by an adverse aggregate or idiosyncratic shock. In their incomplete-markets framework, that service flow generates a convenience yield because the holder can obtain cash without liquidating at depressed prices. Safe assets therefore tend to pay off in states in which marginal utility is high. That perspective is directly relevant for reserve management, where institutions are often especially reluctant to realize mark-to-market losses in stress episodes. The mechanism studied here uses collateralized liquidity provision to reproduce part of those liquidity services at the level of the reserve claim, even though the underlying portfolio is more diversified than a conventional safe-asset portfolio.

Finally, the paper is related to the literature on the inefficiencies of decentralized self-insurance. Das et al. (2022) study non-U.S. central banks that accumulate dollar reserves in order to act as *ex post* lenders of last resort to domestic firms with currency mismatches. They show that decentralized reserve accumulation can generate a macro-financial externality: individual central banks do not internalize the effect of reserve demand on global safe rates and, through that channel, on private-sector incentives. By shifting part of the liquidity and diversification problem from individual central-bank balance sheets to a pooled vehicle, the SRD provides a cooperative institutional arrangement that can reduce the need for decentralized reserve accumulation or the Pigouvian taxes that are optimal in that model.

3. The Economics of Self-Insurance

The current architecture of the Global Financial Safety Net creates a wedge between the private return to reserve diversification and the social value of reserve insurance. For many emerging-market and developing economies, reserves remain the main layer of external insurance that is both pre-positioned and immediately deployable. The evidence reviewed in this section establishes three points. First, reserve accumulation is associated with greater resilience in periods of market stress. Second, that reliance on reserves reflects the fragmented and state-contingent nature of the broader GFSN. Third, when many countries respond in this way, the result is both a quasi-fiscal burden for reserve holders and a broader safe-asset-scarcity externality. Put differently, self-insurance is constrained optimal at the country level but inefficient at the system level.

In itself, the fact that demand for safe and liquid assets (e.g. for reserve accumulation) puts downward pressure on the return of such assets is not in itself problematic or a reason for intervention.[5] However, the literature on safe asset scarcity or the global financial cycle point to equilibrium consequences of these pecuniary externalities, such as excessive leverage (Mendoza and Quadrini, 2024) or FX mismatches (Das et al., 2022), or from low aggregate demand and a Safe Asset trap environment which lowers global output (Fornaro and Romei, 2019). Benigno, Fornaro, and Wolf (2025) similarly point to a related externality through which capital inflows stimulate the US nontradable sector and tend to shrink the more innovative tradable sector, curbing global growth. While the literature's response to these issues tends to be that a Pigouvian tax on safe and liquid assets can resolve the externality and restore constrained efficiency, this paper (and related conceptual work foreshadowed in Section 8) suggests attacking the problem from the other side of the market by making the supply of safe and liquid assets more elastic, thus mitigating the impact of (self-)insurance demand on the safe rate.

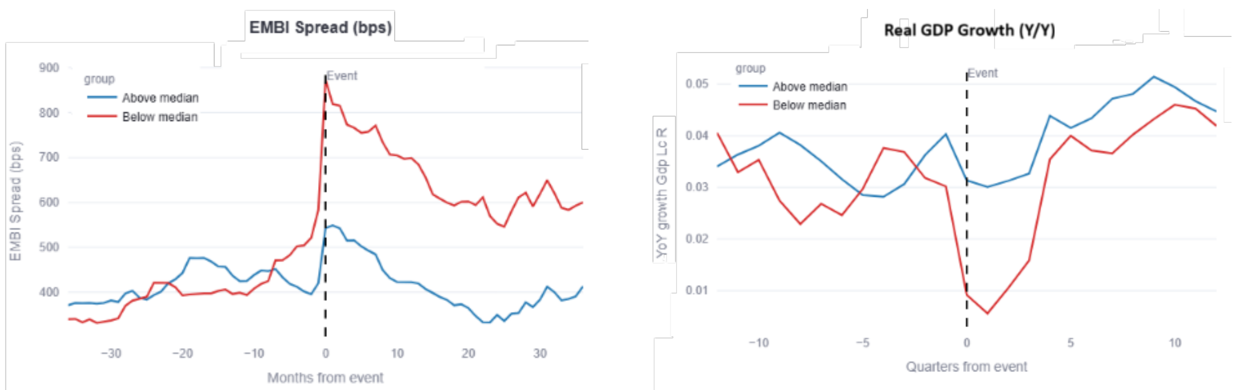
3.1 The Insurance Value of Reserve Accumulation

Despite their opportunity cost, international reserves remain a central instrument of external self-insurance. Because reserves are fully controlled by the sovereign and can be mobilized immediately, they provide a balance-sheet buffer against external stress. That buffer can compress sovereign risk premia when financing conditions tighten, reduce the need for abrupt macroeconomic adjustment, and lower the probability that a liquidity shock turns into a broader balance-of-payments crisis. Empirical evidence supports this interpretation. Gourinchas and Obstfeld (2012) show that, for emerging markets, higher foreign-exchange reserves significantly predict a lower probability of subsequent financial crises. Event-study evidence also suggests that reserves matter for the severity of stress episodes once they occur. Following Acalin et al. (2026), market-stress episodes can be defined as periods in which a country's EMBI spread rises more than 1.5 standard deviations above its historical mean; countries can then be sorted by cumulative reserve accumulation from ten quarters before the event to three quarters after it, scaled by pre-event GDP. In those episodes, above-median reserve accumulators exhibit materially lower sovereign spreads and smaller output losses than below-median accumulators. In the underlying estimates, the spread differential remains substantial for up to two years after the onset of stress, and GDP growth at the trough is roughly 2 percentage points higher for the high-accumulation

[5] In models in which asset-holding decisions are in line with relative risk preferences and there are no other externalities, the accumulation of reserves by Central Banks can be globally efficient (Maggiori, 2017; Gourinchas and Rey, 2022)

group.

Figure 1: Sovereign spreads and GDP growth around market stress episodes.



Note: Financial stress events correspond to periods in which a country's EMBI spread rises more than 1.5 standard deviations above its historical mean. When several events occur within two years, only the first is retained. Countries are divided evenly according to cumulative reserve accumulation, defined as the change in reserves between ten quarters before the event and three quarters after it, scaled by GDP at $t - 10$. The median change in reserves across all events is 0.5 percent of GDP, with medians of 2.5 percent for the above-median group and -1.0 percent for the below-median group. The figure plots median responses for each group. Sovereign spreads are measured in basis points, and GDP growth refers to year-on-year changes. Source: Acalin, Aguiar, Fourakis, Gutiérrez, and Peralta-Alva (2026).

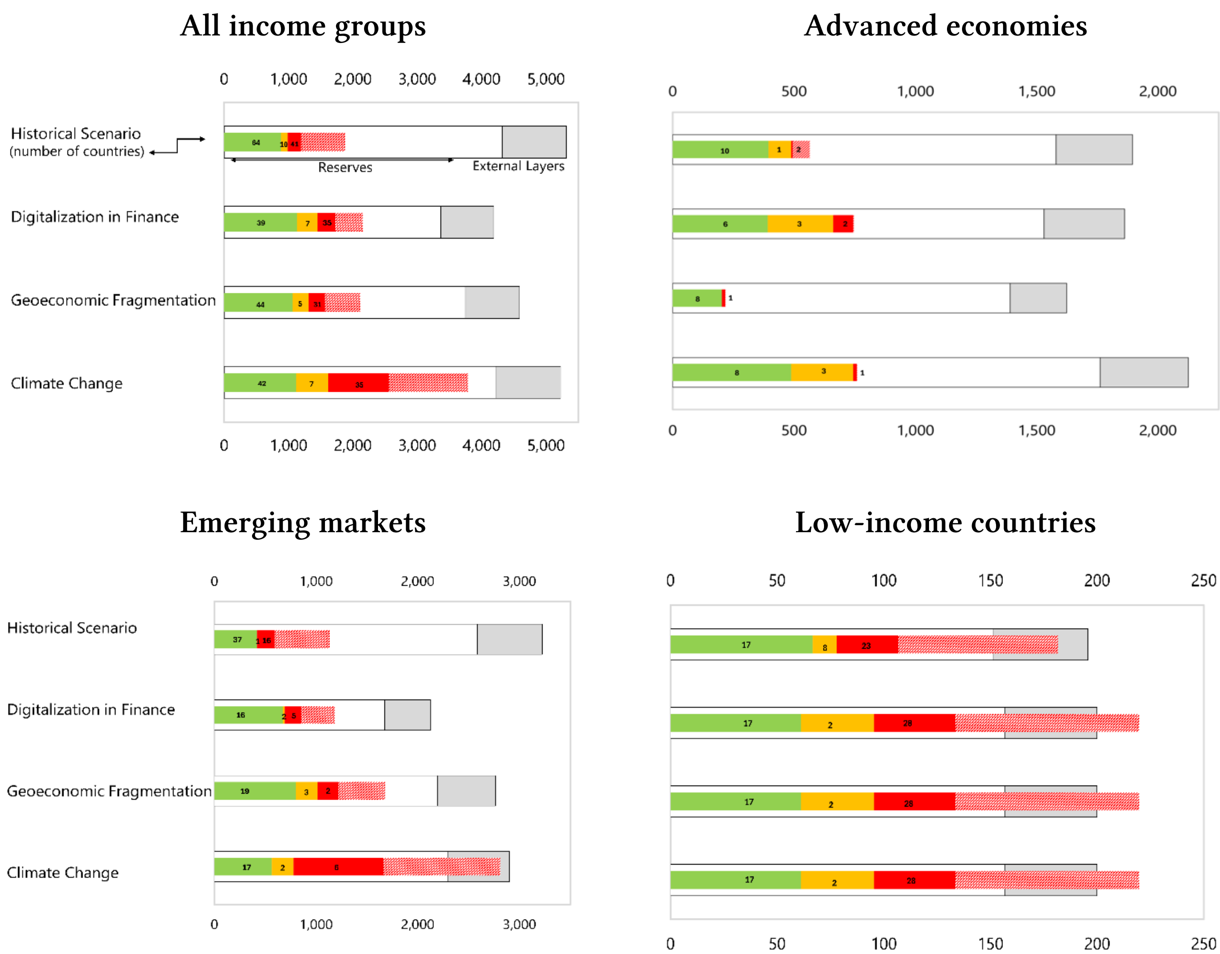
3.2 *Structural Fragmentation of the Global Financial Safety Net*

If reserves are effective, why is their decentralized accumulation a macroeconomic problem? The answer lies in the structure of the broader GFSN. The safety net is often described as having four layers: international reserves, bilateral swap arrangements (BSAs), regional financing arrangements (RFAs), and IMF resources. In practice, however, these layers are not equivalent from the standpoint of a country facing stress. Some instruments create new international liquidity when activated, whereas others mainly redistribute existing liquidity across members. Access conditions also differ sharply across users, currencies, and states of the world.

Country-level protection is therefore determined by the set of resources that can actually be mobilized by a specific country in a given state of the world rather than by global aggregates. A small set of economies enjoys standing or highly reliable access to large bilateral swap lines. For most EMDEs, in contrast, the relevant resource envelope at the onset of a crisis consists of usable reserves, IMF access within ordinary limits, and any RFA or BSA support available under instrument-specific caps and eligibility conditions. Because coordination across layers is limited and activation is often sequential, published aggregate capacities cannot be interpreted as a single pool available on identical terms in crisis times.

Recent IMF stress-scenario analysis reinforces this point (International Monetary Fund, 2023). When country-specific financing needs under stress are compared with the resources each country can plausibly mobilize, aggregate global resources may appear sufficient even though country-level adequacy breaks down. Shortfalls are especially pervasive for low-income countries and remain sizable in dollar terms for emerging markets.

Figure 2: Scenario analysis: demand and available supply—country-level view *(in U.S. dollars, billions)*



Note: The bars decompose stressed financing needs into four categories: countries that can cover needs using reserves only; countries that need both reserves and external layers to meet needs; countries with insufficient coverage for the portion met by the GFSN; and countries with insufficient coverage for the portion not met by the GFSN. Amounts exclude the external financing needs of reserve issuers, although spillovers from those economies are embedded in the simulation. Source: IMF staff calculations; see IMF, *The Global Financial Safety Net—A Stocktaking (2023).*

3.3 *The Opportunity Cost of Reserves and the Safe-Asset Scarcity Externality*

The insurance value of reserves and the fragmentation of external backstops induce EMDEs to rely on self-insurance. Because reserve assets must remain immediately usable under stress, decentralized reserve management continues to concentrate precautionary demand in short-duration safe assets. Holding reserves accordingly entails an opportunity cost. A standard quasi-fiscal measure of these costs is

$$\Phi_{i,t}^{QF} = X_{i,t}\left(r_{i,t}^{dom,T} - r_t^{f,\ell}\right), \tag{1}$$

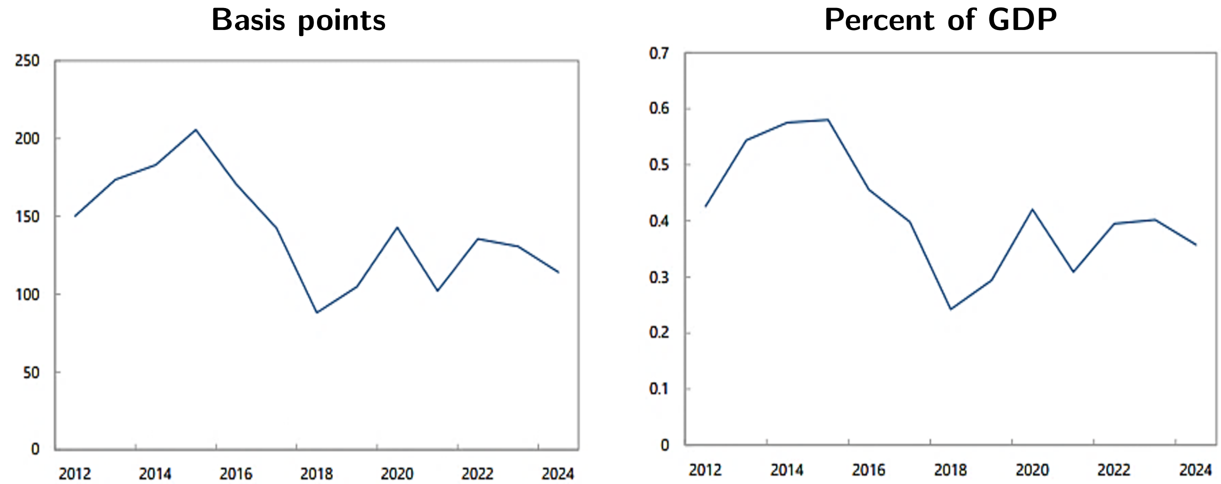
where $X_{i,t}$ is the stock of reserves of country i, $r_{i,t}^{dom,T}$ is the sovereign funding cost at representative maturity T, and $r_t^{f,\ell}$ is the return on short-dated safe reserve assets. This wedge can be decomposed as

$$r_{i,t}^{dom,T} - r_t^{f,\ell} = \underbrace{\left(r_{i,t}^{dom,T} - r_t^{*,T}\right)}_{\text{sovereign spread}} + \underbrace{\left(r_t^{*,T} - r_t^{f,\ell}\right)}_{\text{term/liquidity premium}}, \tag{2}$$

where $r_t^{*,T}$ denotes the yield on a reserve-currency sovereign at maturity T. The first term on the right reflects the sovereign spread for country i, while the second term is a term/liquidity premium.

Figure 3 reports the estimated cost of holding foreign-exchange reserves for emerging-market economies. The left panel shows the cost in basis points, while the right panel scales the cost by GDP.

Figure 3: Cost of Holding FX Reserves for EMs.



Sources: IMF staff calculations based on Bloomberg, World Bank, U.S. Treasury Department, and IFS. *Notes:* The cost of holding reserves for each country is estimated as the difference between the U.S. treasury yield at its average debt maturity and the average maturity of foreign-held Treasuries, plus a country-specific sovereign spread estimated by IMF staff. The aggregate EM index is weighted by the reserve holdings of each country.

The ultimate objective of the SRD is to allow countries to obtain a better interest rate r_t^{pool} (while retaining the safety and liquidity properties of reserve assets) in order to reduce this quasi-fiscal cost, by the amount $(r_t^{pool} - r_t^{f,l})X_{i,t}$.

When the supply of safe assets is inelastic, aggregate decentralized reserve demand, $\sum_i X_{i,t}$, raises the convenience premium on short-duration safe assets and compresses their yields (Caballero, Farhi, and Gourinchas, 2017). This increases the opportunity cost of self-insurance under both the standard quasi-fiscal metric and the shadow-liquidity-premium metric. As argued by Das et al. (2022), these effects may also tighten global lower-bound constraints and encourage risk-taking elsewhere in the system. The reserve-management problem is therefore linked to a broader pecuniary externality.

These observations define the economic problem studied in the remainder of the paper. The challenge is not merely to maximize yield while maintaining low risk levels. It is to preserve immediate usability while mitigating the system's incentive to concentrate precautionary demand in the narrowest class of safe liquid assets. The SRD operates precisely on that margin: it embeds the liquidity function within the pooled structure that backs the reserve claim, rather than leaving each reserve holder to solve the diversification problem independently on its own balance sheet.

4. Balance Sheet Mechanics

This section describes the SRD in balance-sheet terms and fixes notation. The benchmark SRD is a floating-value claim on a pooled portfolio. In this scheme, all market risk is borne by SRD holders. A participant that converts units into cash receives the contemporaneous realizable market value of those units. The pooled vehicle may hold an internal cash buffer, but that buffer affects settlement and execution costs rather than the redemption price or the aggregate quantity of redemptions. A standing collateralized borrowing window may be layered on top of the claim to support short-horizon liquidity needs without forcing immediate liquidation. This also allows holders in need of liquidity to remain exposed to the assets underlying the SRD and "ride" temporary drops in asset values.

The architecture is deliberately close to familiar reserve-management practice. Central banks already separate reserves into working-capital, liquidity, and investment tranches. The SRD does not replace that logic; it aggregates it. The settlement buffer corresponds to the liquidity function, the diversified portfolio corresponds to the investment function, and the collateralized borrowing window, where present, provides a rule-based way to convert the pooled claim into foreign-currency cash without immediate asset liquidation. The novelty is therefore not the existence of tranches, but the relocation of tranche management from each individual reserve holder to a pooled balance sheet.

It is useful to distinguish three balance sheets: that of the participant, that of the pooled vehicle, and, when secured borrowing is used, that of the liquidity provider. The borrowing window is optional. It allows a participant to obtain cash without immediately redeeming units, but it is not part of the baseline valuation rule.

4.1 Claim Structure and Valuation

At date t, participant i holds $q_{i,t}$ SRD units. Let

$$Q_t = \sum_i q_{i,t} \tag{3}$$

denote the number of units outstanding. The pooled vehicle holds two asset categories,

$$A_t = C_t + I_t, \tag{4}$$

where C_t is the value of cash and near-cash assets held for settlement purposes and I_t is the contemporaneous market value of the investment portfolio. In the benchmark exposition, the pooled vehicle has no external leverage, no junior capital layer, and no principal guarantee. Its net asset value therefore equals the value of its assets,

$$V_t = A_t. \tag{5}$$

The unit value is

$$v_t = \frac{V_t}{Q_t} = \frac{A_t}{Q_t}, \tag{6}$$

and the marked-to-market value of participant i's position is

$$S_{i,t} = q_{i,t} v_t. \tag{7}$$

For the purposes of this paper, reserve usability refers to the participant's ability to obtain foreign-currency cash promptly under a pre-specified rule. In the benchmark design, that rule is redemption at contemporaneous executable value.

4.2 Issuance and Redemption at Contemporaneous Market Value

At issuance, participant i contributes reserve assets and receives SRD units of equal market value. On the participant's balance sheet, the transaction is an asset swap: directly held reserve assets are replaced by a claim on the pooled vehicle. The cash buffer and the investment portfolio are therefore internal components of the pooled balance sheet rather than separately managed portfolios on each participant's balance sheet.

Let $m_{i,t}$ denote the number of units redeemed by participant i at date t, and let

$$M_t = \sum_i m_{i,t} \tag{8}$$

denote aggregate redemptions. The only aggregate quantity restriction is

$$0 \leq M_t \leq Q_t. \tag{9}$$

There is no separate cap on aggregate redemptions tied to the size of the cash buffer.

To state the redemption rule precisely, let $V_t^{\text{exec}}(M_t)$ denote the net asset value of the pooled vehicle after the transactions required to meet aggregate redemptions M_t have been executed at contemporaneous market prices, but before cash is distributed to redeemers. The corresponding common execution value per unit is

$$v_t^{\text{exec}}(M_t) = \frac{V_t^{\text{exec}}(M_t)}{Q_t}. \tag{10}$$

Participant i then receives

$$p_{i,t} = m_{i,t} v_t^{\text{exec}}(M_t), \tag{11}$$

and aggregate cash paid to redeemers is

$$P_t = M_t v_t^{\text{exec}}(M_t). \tag{12}$$

This formulation excludes redemptions at stale values or at values insulated from liquidation losses. If meeting P_t requires asset sales, the prices obtained on those sales enter $V_t^{\text{exec}}(M_t)$. Redeemers therefore receive the contemporaneous realizable value of the redeemed claim rather than a protected value based on pre-sale accounting prices.

The internal cash buffer affects implementation but not the valuation rule. If $P_t \leq C_t$, redemptions can be settled from cash on hand. If $P_t > C_t$, the pooled vehicle liquidates assets and transfers the proceeds to redeemers. In either case, redeemed and non-redeemed units are valued on the same basis. After the redemption is settled, the value of the remaining pool is

$$V_t' = V_t^{\text{exec}}(M_t) - P_t = (Q_t - M_t)v_t^{\text{exec}}(M_t). \tag{13}$$

The cash buffer is therefore a settlement buffer and a device for reducing expected transaction costs in ordinary states. It does not create a senior redemption claim and does not protect early redeemers from market losses.

Reserve-asset recognition. The analysis treats reserve eligibility as a design constraint, not as an automatic implication of pooling. In practice, SRD units could be counted as reserves only to the extent that the units themselves satisfy the relevant reserve-asset criteria. A minimum standard is to establish redemption or liquidation at contemporaneous executable value within a few business days which the SRD satisfies. However reserve status also depends on country specific legal and operational constraints which will need to be considered.[6]

The statistical recording of SRD units would depend on the final legal form of the vehicle. A mutual-fund-like claim may be recorded as an investment-fund share or equity security even if the underlying assets are fixed-income instruments. That classification need not mechanically preclude reserve-asset treatment, but it may interact with domestic reserve-management laws and internal policies that restrict equity or fund-share holdings. These legal and accounting questions are therefore implementation constraints for country participation, not parameters determined by the quantitative exercise.

4.3 *Optional Standing Borrowing Window*

A participant may prefer not to redeem units in states in which immediate liquidation is expensive. The design may therefore include a standing collateralized borrowing window against SRD units. To use this facility, a participant would "post" $S_{i,t}^{\text{col}} \leq S_{i,t}$ as collateral (with a constraint

[6] The ability to raise funds by pledging the units as collateral does not by itself make the units reserve assets. Likewise, any portion of a participant's SRD holdings that is pledged or otherwise encumbered would not be treated as freely available reserve assets while the encumbrance remains. The liquidity obtained from the advance may be immediately usable foreign exchange but the pledged SRD units are not.

that only the market value of the current holdings $S_{i,t}$ can be pledged). Let $B_{i,t}$ denote the advance and $\tau_{i,t}$ its maturity in days. The borrowing constraint is

$$B_{i,t} \leq \min\left\{(1-h_t)S_{i,t}^{\text{col}},\ \bar{B}_{i,t}\right\}, \tag{14}$$

where h_t is the haircut and $\bar{B}_{i,t}$ is the participant-specific borrowing limit. If the posted window rate is i_t^W, repayment due at maturity is

$$D_{i,t+\tau_{i,t}} = B_{i,t}\left(1 + i_t^W \frac{\tau_{i,t}}{360}\right). \tag{15}$$

The intended institutional analogy is a discount-window-style facility: short maturity, pre-specified pricing, conservative haircuts, and margining.

This transaction does not alter the redemption rule. The participant chooses between redeeming units, borrowing against units, or combining the two. Borrowing changes the timing of liquidation. It does not change the allocation of market risk. If the value of the underlying portfolio falls, that loss remains with the holder of the SRD claim. While units are pledged, the encumbered portion is not part of the holder's freely available reserve assets. The liquidity obtained through the advance is the immediately usable foreign-currency asset.

4.4 *Margining and the Role of the Public Layer*

If the borrowing window is used, short-horizon collateral management relies on rules-based adjustments to avoid forced liquidation. Let

$$G_{i,t+1} = \max\left\{0,\ B_{i,t} - (1-h_{t+1})S_{i,t+1}^{\text{col}}\right\} \tag{16}$$

denote the collateral gap at the next valuation date. If $G_{i,t+1} > 0$, coverage is restored through one or more rules-based actions: additional eligible collateral is posted, part of the advance is repaid, or the pledged position is reduced. Only if these adjustments fail does forced liquidation arise within the lending facility.

The role of a public or quasi-public backstop is strictly limited to providing short-maturity advances against pre-positioned SRD collateral. It is not needed to sustain unit value, to guarantee principal, or to make redemptions possible, as redemptions are already feasible through the pooled vehicle at contemporaneous realizable value. When a participant chooses secured borrowing, the public layer provides these advances at a posted rate, subject to conservative haircuts, access caps, and margining.

Providing this collateralized liquidity through a public backstop rather than private dealers avoids critical informational and cyclical frictions. If private dealers active in related sovereign or funding markets supply the credit line, a participant's request for liquidity may reveal information about stress or portfolio needs. In addition, private funding capacity may contract precisely in the states in which reserve liquidity is most valuable. A public backstop, if present, therefore serves exclusively as a contingent source of collateralized liquidity to bypass these vulnerabilities. It does not absorb ordinary market losses on the pooled portfolio.

5. Quantitative Illustrations

The International Monetary Fund (2013) reserve-management guidelines recommend securing foreign-exchange liquidity to meet short-term needs under stress scenarios before maximizing risk-adjusted returns over the medium to long term. Central banks typically implement this mandate through a tranching framework. The 2023 World Bank Reserve Advisory and Management Partnership (RAMP) survey documents that central banks size reserve tranches using reserve-adequacy metrics, most commonly short-term liquidity needs, including potential FX interventions, followed by import coverage, short-term external debt relative to reserves, the IMF's ARA metric, and broad money. The survey also reports that 62 percent of respondents use more than one metric when setting tranche sizes, consistent with the multiple objectives assigned to working-capital, liquidity, and investment tranches. Investment tranches are managed over longer horizons, averaging 44 months compared with 17 months for liquidity tranches (World Bank, 2023).

The benchmark portfolio we consider contains only a subset of US dollar-denominated instruments: US Treasuries, highly rated (A grade or higher) sovereign and sub-sovereign bonds, investment-grade corporate bonds, and two equity indices (the MSCI World and S&P 500). This benchmark portfolio is used as an illustration of the benefits of the SRD scheme. The input data covers the period between January 2003 and December 2024. This asset menu mirrors the contemporary composition of central bank reserve portfolios. While government bonds and sovereign, supranational, and agency (SSA) securities constitute the core of reserve holdings, the RAMP survey identifies investment-grade corporate bonds as the most prevalent nontraditional asset class. Allocations to developed-market equities remain comparatively modest: only 22 percent of central banks permit equity investments, with an average allocation of 1.5 percent unconditionally and 5.4 percent among those with active exposure (World Bank, 2023). We therefore impose a 20 percent upper bound on the combined equity share. This cap serves as a permissive constraint to explore the risk-return frontier rather than a description of average empirical holdings.

To construct the benchmark investment allocation, we simulate a large set of candidate portfolios. Let $w_p = (w_{p1}, \dots, w_{pN})'$ denote the vector of portfolio weights for candidate p. To guarantee a baseline allocation to safe, short-maturity sovereign assets, we assign a fixed weight of 0.10 to the 1–3-year US Treasury index in every candidate portfolio. We distribute the remaining

0.90 across assets 2 through N using independent draws from a standard gamma distribution, normalized to sum to 0.90. For performance benchmarking, we independently generate a set of Treasury-only portfolios restricted exclusively to indices of Treasury assets.

We apply a strict set of admissibility criteria to evaluate candidate allocations based on downside risk and expected return relative to the Treasury-only benchmark. A portfolio is retained if and only if it satisfies three constraints: (i) the combined weight on the MSCI World and S&P 500 indices does not exceed 0.20; (ii) the fifth percentile of its annualized rolling 36-month returns strictly exceeds the corresponding median value from the Treasury-only benchmarks; and (iii) its mean annualized return exceeds the Treasury-benchmark median. The 36-month rolling window explicitly aligns our downside-risk constraint with the medium-term horizon typical of reserve investment tranches. Among the admissible portfolios, we select the benchmark allocation, denoted w^*, that maximizes the mean annualized return. Appendix B provides a comprehensive formalization of the simulation and selection algorithm.

The Treasury-only benchmark is similarly built, but it includes only indices of U.S. Treasuries of different maturities. The share allocated to 1-3 year maturies is fixed at 10 percent, and the rest is allocated to indices including 1–5 and 1–10 year maturities. The full details can be found in the appendix.

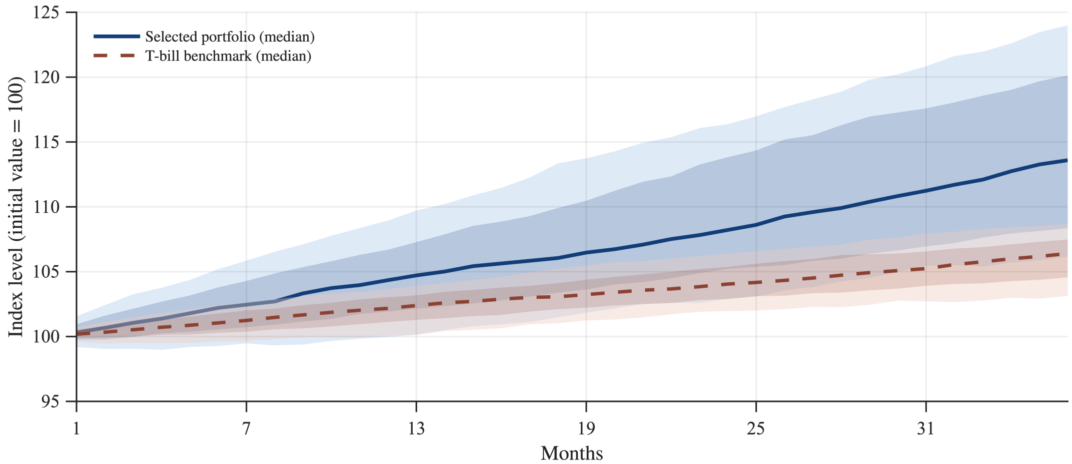
Finally, it is important to note that nothing in the SRD is specific to the U.S. dollar. The same balance-sheet logic could be applied to multi-currency portfolios or to a portfolio benchmarked to the SDR basket, with currency composition treated as an additional design choice.

5.1 *Risk-Return Properties Relative to Traditional Reserves*

Figure 4 compares the distribution of simulated 36-month index paths for the benchmark portfolio and the T-bill benchmark. In this calibration, the fan chart shows a clear upward shift in the distribution of the benchmark portfolio relative to the T-bill benchmark. In the benchmark calibration, the benchmark portfolio has an average annualized 36-month return of about 4.5 percent, compared with about 2 percent for the T-bill benchmark.

The dispersion of outcomes is naturally larger for the benchmark portfolio, but the downside remains contained. The Figure makes it clear that the benchmark portfolio has higher volatility than the T-bill benchmark. However, its excess return quickly dominates: after about 20 months, the 25th percentile of returns under the benchmark portfolio is already above the median return of the T-bill benchmark; by the end of the 36-month horizon, the 25th percentile of returns for the benchmark portfolio surpasses the 75th percentile of returns for the T-bill benchmark. The 5th percentile of annualized 36-month returns is 1.67 percent for the benchmark portfolio, compared with 0.70 percent for the T-bill benchmark. At the monthly frequency, the benchmark

Figure 4: Simulated portfolio paths: benchmark portfolio vs. T-bill benchmark.



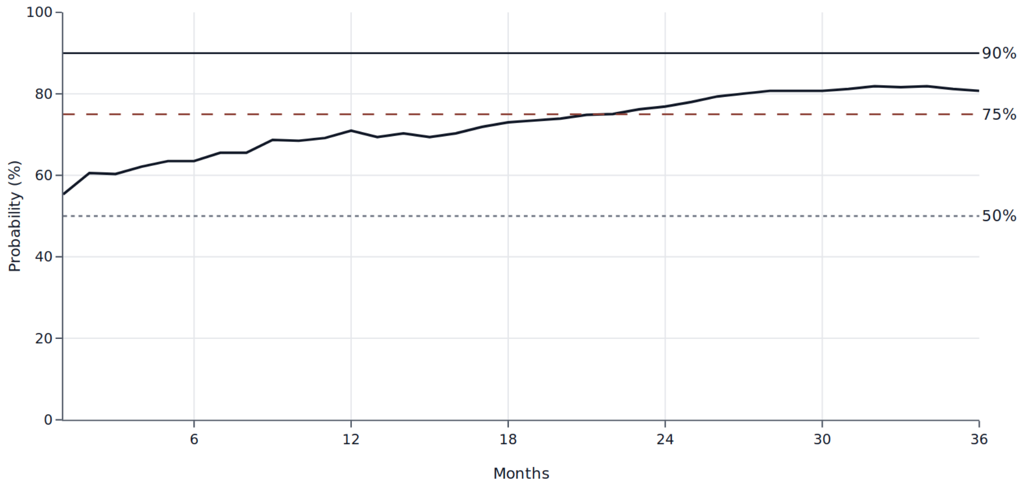
Note: The figure reports simulated valuation paths over a 36-month horizon. Shaded areas correspond to the 10th–90th and 25th–75th percentile bands across simulations, while solid lines denote median (50th percentile) outcomes. The underlying return distribution is generated using the hybrid semi-parametric simulation described in Appendix B.

portfolio has an annualized return of 4.59 percent and annualized volatility of 3.36 percent, while the benchmark delivers 2.00 percent with 1.52 percent volatility. Monthly Value-at-Risk at the 5 percent level is −1.21 percent for the benchmark portfolio, compared with −0.53 percent for the benchmark. Taken together, these results indicate that the benchmark portfolio improves substantially over the T-bill benchmark in terms of expected return, while keeping downside risk at a moderate level. In other words, while the benchmark portfolio does display a higher VaR, its increased returns only need a few years to shift the entire distribution above that of the T-bill benchmark. The simulation procedure underlying these results is described in Appendix B.

To assess the timing of outperformance, Figure 5 reports, for each investment horizon, the empirical probability that the benchmark portfolio beats the T-bill benchmark. The exercise is computed from the simulated return paths by normalizing both the benchmark portfolio and the benchmark indices to 100 at inception and then calculating, at each month, the fraction of paths in which the portfolio index is above the benchmark index. The figure shows that the probability of outperformance is already above 50 percent at short horizons, starts at approximately 55 percent, and rises steadily over time. It crosses the 75 percent threshold around month 23 and stabilizes close to 80–82 percent toward the end of the 36-month horizon, while remaining below the 90 percent threshold. This pattern indicates that the benchmark portfolio's advantage becomes increasingly robust as the investment horizon lengthens.

These figures should be interpreted as calibration-specific illustrations rather than estimates of attainable SRD returns. The numerical levels depend on the asset menu, sample period, de-

Figure 5: Time to Outperform the T-Bill Benchmark



Note: The figure reports the empirical probability that the benchmark portfolio outperforms the T-bill benchmark at each horizon month. Probabilities are computed as the share of simulated paths in which the portfolio index exceeds the benchmark index, with both indices normalized to 100 at inception.

pendence structure, and 36-month evaluation horizon. The relevant point is therefore not the precise return differential, but the possibility that a reserve-compatible pooled portfolio can improve expected returns while satisfying explicit downside-risk constraints.

The benchmark portfolio may underperform the T-bill benchmark in adverse states of the world, as any portfolio that takes duration, credit, and equity exposure can. But the risks are calibrated by design: criterion (ii) limits the portfolio's ex ante risk while allowing the SRD to earn a higher expected return than a purely short-term UST allocation.[7]

5.2 *Asset Pricing Interpretation*

A critical metric of any portfolio is how returns co-vary with states of the world in which global liquidity is scarce. To evaluate this margin, we estimate a stochastic discount factor (SDF), $\hat{M}_t$, which summarizes state prices across time. The SDF is constructed following standard asset pricing methods as a linear combination of traded basis assets (see Appendix C for more details). Let $\hat{M}_t$ denote the estimated SDF, where higher values of $\hat{M}_t$ correspond to adverse states characterized by high marginal utility. We estimate the reduced-form relationship

$$R_t^j = \alpha_0^j + \beta_M^j \hat{M}_t + \varepsilon_t^j. \tag{17}$$

[7] Figure 17 shows the performance of the benchmark portfolio during the tightening cycle leading to the SVB episode in 2022–23.

Figure 6: Monthly Returns (%) of the S&P 500, the Selected Investment Portfolio, and 1–3-Year US Treasuries Plotted Against the Estimated SDF.

Notes: Higher SDF values correspond to worse aggregate states. Fitted lines show a strongly negative slope for equities and flat or slightly positive slopes for the selected investment portfolio and short Treasuries. Source: IMF staff calculations based on Haver Analytics.

where R_t^j denotes the monthly return on asset or portfolio j, with

$$j \in \{\text{benchmark portfolio (SRD)}, \text{S\&P 500}, \text{1–3-year U.S. Treasuries}\}$$

. We estimate this regression separately for each j. A negative loading on $\hat{M}_t$ indicates that the asset performs poorly in bad states and therefore commands a higher risk premium, while a positive loading indicates that the asset provides insurance in adverse states, in the spirit of Brunnermeier, Merkel, and Sannikov (2024).

In our sample, the selected investment portfolio exhibits a near-zero to slightly positive loading on the estimated SDF ($\beta_M^j \geq 0$), behaving much closer to short-duration Treasuries than to a portfolio of equities. By contrast, pure equity returns load heavily and negatively on the SDF. This distinction indicates that our diversified, fixed-income-dominant portfolio does not simply inherit the procyclical properties of its equity sleeve, retaining favorable payoff characteristics in bad states.

Figure 6 illustrates this graphically, plotting the monthly returns of the S&P 500, the selected investment portfolio, and 1–3-year US Treasuries against the estimated SDF. Equities exhibit a steep negative slope, meaning that their lowest returns occur when resources are more valuable. Short Treasuries covary positively with the SDF, and while the selected portfolio displays a slight negative correlation with the SDF, it is much milder than that of equities, and it also obtains a higher average level than the portfolio of Treasuries.

6. Liquidity Frictions and Tail Risk

The gains found so far can be supplemented by a discount-window-like facility. Such a facility would allow participants to borrowing short-term against their share in the scheme. This would allow participants to avoid having to sell when asset prices are low but expected to revert. This section describes haircut and margining practices which would allow such borrowing to be undertaken without risk to the overall scheme, because residual risks would remain with SRD holders.

Using the high-frequency daily simulated portfolio returns generated by the hybrid semi-parametric model detailed in Appendix B, we analyze collateral dynamics under stress. Rather than estimating a structural model of crisis dynamics, our objective is to characterize the empirical interaction among valuation losses, collateral haircuts, and margining over the brief horizon relevant for emergency liquidity provision.

6.1 Haircuts, Margin Calls, and Residual Exposure

As described earlier, the contractual terms that matter for a collateralized liquidity operation are the pledged collateral $S^{\text{col}}_{i,t}$, the advance $B_{i,t}$, the haircut h_t, the maturity $\tau_{i,t}$, and the margin rule. The simulation below takes these terms as given and asks how often the realized path of collateral value would exhaust the haircut buffer over the life of the loan.

Consider an advance originated at date t with maturity $\tau = 30$ days. For a given haircut h, the cash advanced at origination is set equal to the lending value of the pledged SRD position,

$$B_{i,t} = (1-h)S^{\text{col}}_{i,t}. \tag{18}$$

The selected investment portfolio from the previous subsection is then used as the collateral pool. Let $R_{t,t+s}$ denote the cumulative simulated return on that portfolio between dates t and $t+s$. Along each simulated path, the marked-to-market value of pledged collateral evolves as

$$S^{\text{col}}_{i,t,t+s} = S^{\text{col}}_{i,t}\left(1 + R_{t,t+s}\right), \qquad s = 1, \dots, 30. \tag{19}$$

The relevant event for the lender is that the realized collateral value falls below the outstanding advance before maturity. This occurs when

$$\min_{1\le s\le 30} S^{\text{col}}_{i,t,t+s} < B_{i,t}, \tag{20}$$

which can be expressed as

$$\min_{1\le s\le 30} R_{t,t+s} < 1-h. \tag{21}$$

The second equation states that this is equivalent to asking whether the simulated cumulated return path crosses the haircut threshold $1-h$ at any point within the 30-day window. Repeating this calculation across simulated paths yields, for each haircut, the cumulative probability that the initial buffer is exhausted before the loan matures.

This is the simulation counterpart of the balance-sheet mechanics in Section 4. The haircut determines the initial distance between collateral value and cash advanced. The daily return process determines how quickly that distance can be reduced. The margin rule then determines the consequences of a breach once it is observed at the next valuation date. Because the collateralized liquidity layer, if present, provides only short-maturity advances against pre-positioned collateral and does not absorb the long-run market risk of the SRD, this short-horizon breach probability is the relevant object for the analysis that follows.

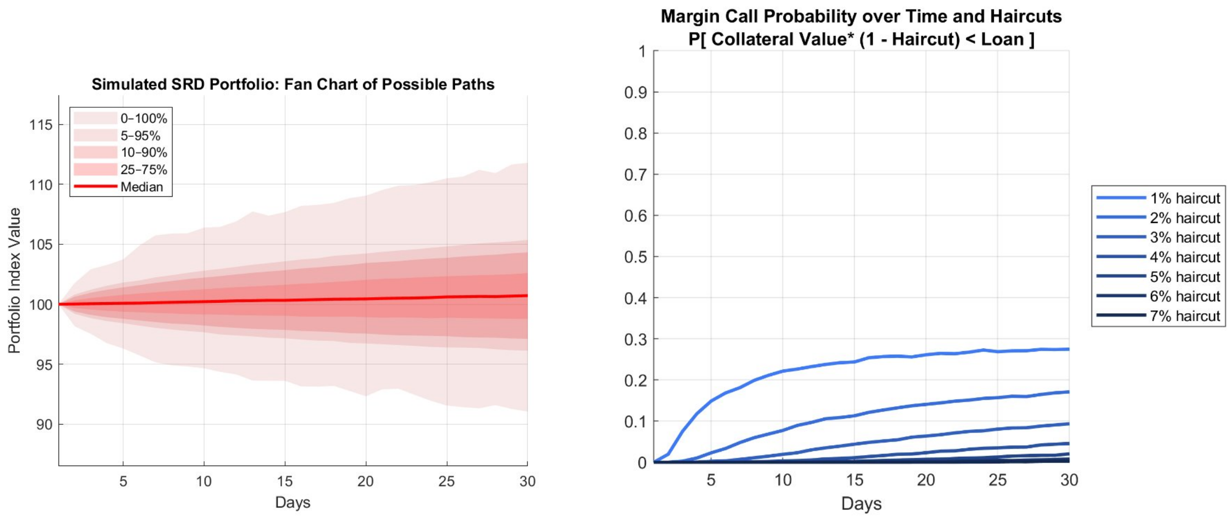
Using our simulated daily return panel, we estimate the cumulative probability that portfolio value breaches the haircut-adjusted collateral threshold over a 30-day horizon. Figure 7 shows the probability distribution for the value of the portfolio over a simulated month on the left panel, along with the probability of a loss large enough to trigger a margin call on the right panel, for different values of the haircut. In this finite simulation design, haircuts in the high single digits are sufficient to virtually eliminate threshold breaches over the 30-day window.[8] These values are calibration anchors rather than operational recommendations; in practice, haircut policy would need to reflect asset composition, valuation frequency, concentration, stress scenarios, and institutional risk tolerance.

These probabilities should not be interpreted as unconditional frequencies for future crisis events. Their purpose is narrower: they show that, under a rules-based framework combining an initial haircut with frequent margin checks, the residual balance-sheet risk from lending against SRD collateral can be kept low. Because positions are marked to market daily and coverage is restored through additional eligible collateral, partial repayment, or collateral substitution, any residual exposure after a large one-day loss would be limited to the interval between valuation and margin restoration.

The analysis we have conducted this far suggests that the SRD can indeed improve the expected return profile of reserves, and conservative collateral design can keep the short-horizon risk of liquidity provision within a range consistent with the haircut and margining assumptions used here.

[8] Note that because the simulation involves a Student-t distribution with full-support, a zero probability of breach is by construction impossible.

FIGURE 7: FAN CHART FOR THE SELECTED INVESTMENT PORTFOLIO AND MARGIN-CALL PROBABILITIES.



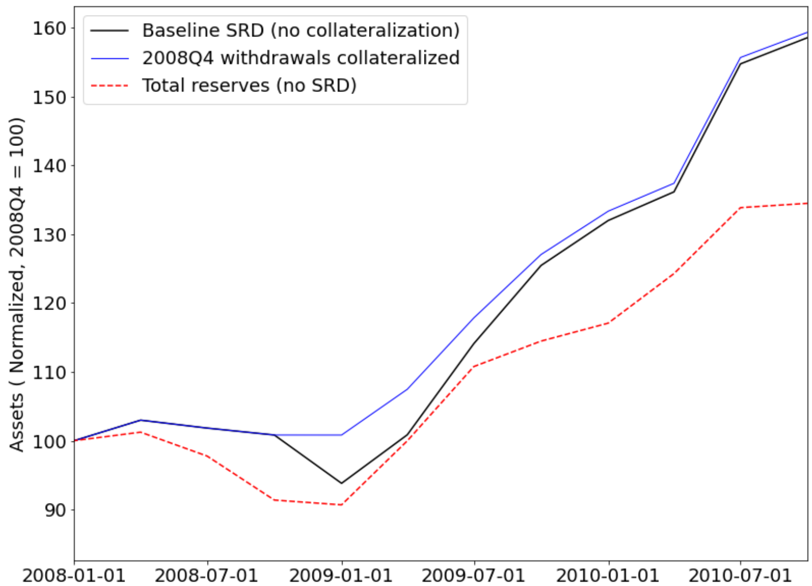
Note: Both panels are based on the hybrid semi-parametric simulation described in Appendix B. The left panel reports percentile bands for 30-day simulated portfolio-value paths. The right panel reports the cumulative share of paths that breach haircut-adjusted collateral thresholds over a 30-day window. Source: IMF staff calculations based on Haver Analytics.

6.2 *An Illustration of the Collateral Facility in a Market Slump*

We consider a counterfactual exercise in which countries are assumed to use a pooled SRD facility. At the start of the simulation, countries deposit their initial reserve holdings into the SRD, which invests the pooled assets in a diversified portfolio with a fixed liquidity buffer. Subsequently, countries withdraw from or replenish their SRD positions in line with observed quarterly changes in reserves from balance-of-payments data, subject to a cap given by each country's share of simulated SRD assets. In the baseline configuration, large withdrawals that exceed the liquidity buffer are met through asset liquidation, generating transaction costs and a discrete decline in the size of the SRD during periods of stress.

We then study a counterfactual in which withdrawals in 2008Q4—the peak quarter of reserve drawdowns during the Global Financial Crisis—are fully collateralized. In this scenario, the SRD does not liquidate assets in that quarter; instead, the withdrawn amounts are treated as a zero-interest loan to the withdrawing countries, which is repaid gradually through subsequent replenishments. By avoiding forced liquidation and associated transaction costs, the SRD preserves a larger investable base, leading to higher asset values in subsequent periods (Figure 8). Under this assumption, the SRD ends up approximately 10% percent larger relative to the baseline after 2 years, which corresponds to an annualized gain of about 4.9 percent. Because no interest or fees are charged on the collateralized portion, this gain should be interpreted as an

Figure 8: SRD evolution with and without collateralization of withdrawals in one period



Notes: The figure shows the evolution of total SRD assets under the baseline simulation (no collateralization), a counterfactual in which withdrawals in 2008Q4 are fully collateralized, and a benchmark corresponding to the total reserves held by the same set of countries in the absence of an SRD. In the collateralization scenario, the SRD does not liquidate assets in the crisis quarter; withdrawn amounts are treated as a zero-interest loan that is repaid through subsequent replenishments. Values are shown relative to the 2008Q1 level.

upper bound on the potential benefits of collateralization, capturing purely the effect of avoiding forced liquidation and the subsequent compounding of returns.

7. Macroeconomic Aggregation and Market Impact

The preceding sections characterize the SRD at the level of a representative pooled balance sheet. This section turns to aggregation. The questions are how large such a structure could become under conservative participation limits, whether that scale corresponds to substantial reserve coverage for participating countries, and how the associated portfolio flows compare with the depth of the underlying asset markets. The exercises are again numerical illustrations: their purpose is to map the cross-country distribution of reserves into plausible subscription capacity and to gage the order of magnitude of any resulting market impact.

7.1 *Demand Estimation and the Subscription Rule*

Let X_i^0 denote the baseline stock of official reserves of country i, $\bar{S}$ a per-country subscription cap to the SRD, and η a reduced-form reserve-demand response to the SRD. Define the effective reserve stock available for subscription as

$$X_i(\eta) = (1+\eta)X_i^0, \qquad \eta \in [\underline{\eta}, \bar{\eta}], \quad \underline{\eta} > -1. \tag{22}$$

An illustrative subscription rule is

$$S_i(\bar{S}, \eta) = \min\left\{(1+\eta)X_i^0, \bar{S}\right\}, \tag{23}$$

where $S_i(\bar{S}, \eta)$ denotes country i's holdings in the SRD. The corresponding coverage ratio, measured relative to the baseline stock of reserves, is

$$\kappa_i(\bar{S}, \eta) = \frac{S_i(\bar{S}, \eta)}{X_i^0} = \min\left\{1+\eta, \frac{\bar{S}}{X_i^0}\right\}. \tag{24}$$

Aggregate assets under management for a given participant set $\mathscr{G}$ are then

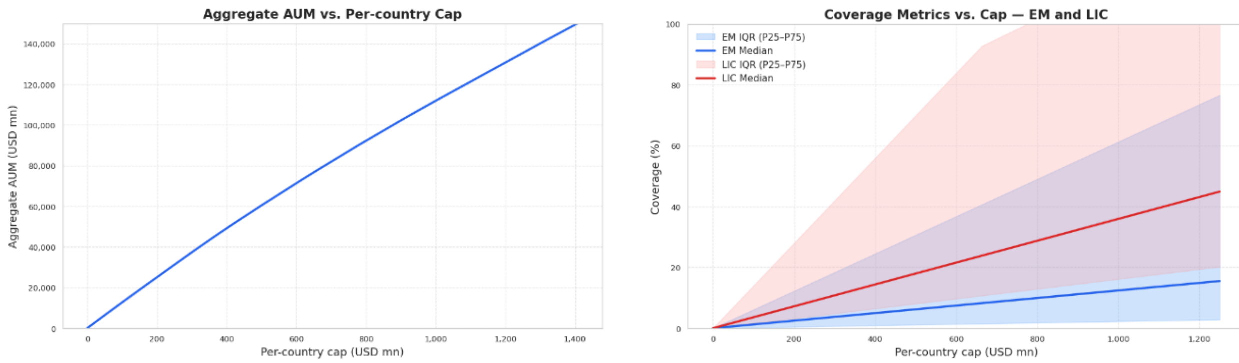
$$\text{AUM}(\bar{S}, \eta) = \sum_{i \in \mathscr{G}} S_i(\bar{S}, \eta). \tag{25}$$

This simple rule captures two aggregation margins. The cap $\bar{S}$ controls concentration and total scale. The parameter η captures the net effect of the SRD on desired reserve holdings. When $\eta = 0$, countries only reallocate existing reserves into the structure. When $\eta > 0$, the lower cost of holding a liquid reserve claim induces additional reserve accumulation. When $\eta < 0$, the SRD reduces desired reserve holdings, for example because improved liquidity insurance or income effects dominate the substitution effect. In the numerical illustrations below, we report the case $\eta = 0.5$.

The calibration uses monthly IMF International Financial Statistics data on official reserves excluding gold for 88 emerging-market economies and 49 low-income countries. The cross section of reserves is highly skewed. That skewness is central to the SRD. A common cap can cover a large share of reserves for smaller reserve holders while keeping total scale modest. In the numerical illustration, a cap of about US$400 million is sufficient for roughly eight countries to place their entire reserve stock, twenty-one countries to place at least one-half, and thirty-eight countries to place at least one-quarter. The structure can therefore provide substantial coverage for a non-negligible set of countries without requiring very large aggregate size.

Figure 9 summarizes this mapping from the cap to aggregate scale and country-level coverage. The left panel shows the increase in aggregate AUM as a function of the cap. The right panel shows the median and interquantile range of the coverage metric (the ratio of potential

Figure 9: Aggregate AUM and Coverage Metrics vs. Per-Country Cap.



Note: EM group includes 88 countries and LIC group 49 countries, based on monthly IMF IFS statistics on official reserves excluding gold (December 2024). Coverage denotes the ratio of each country's potential subscription to its reserves under a given cap. Shaded areas show the interquartile range (P25–P75) of coverage; solid lines indicate medians. Aggregate AUM refers to the sum of subscriptions under each per-country cap scenario. Source: IMF staff calculations.

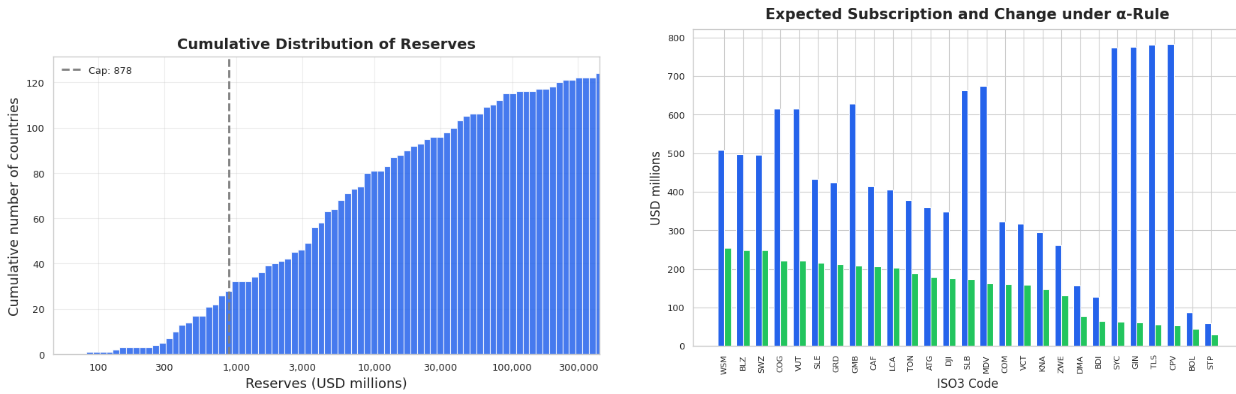
subscription to current reserves). Because reserve holdings are highly skewed, the same cap has very different implications across countries, as can be seen from the wide interquantile ranges shown. For economies with relatively small reserve stocks, even a moderate cap can cover a substantial share of reserves, whereas for larger reserve holders the cap remains a binding constraint on participation.

7.2 *Illustrative Aggregate Scale*

The mapping from the cap to total AUM is mechanical once the reserve distribution is fixed. In the baseline calibration, a per-country cap of about US$408 million yields aggregate AUM of roughly US$50 billion. A larger calibration with total AUM of US$100 billion corresponds to a cap just below US$900 million, a scale consistent with roughly twenty-seven countries placing large shares of their reserves in the structure. These magnitudes are not intended as targets. Their role is to provide reference points: one at which participation is already substantial for a number of reserve holders, and another at which the structure is materially larger while still modest relative to the markets in which the underlying portfolio would be invested.

The effective scale can exceed a pure reallocation of existing reserves if lower quasi-fiscal costs induce some countries to accumulate additional reserves. This is the role of the η-rule. Under $\eta = 0.5$, subscriptions can reach up to 1.5 times current reserves, with the excess interpreted as newly accumulated reserves. The left panel of Figure 10 shows the cross-country distribution of reserve stocks. The right panel maps that distribution into expected subscriptions under the

FIGURE 10: RESERVES DISTRIBUTION AND EFFECTIVE CAP.



NOTE: RESERVE DATA COVER 88 EMERGING-MARKET ECONOMIES AND 49 LOW-INCOME COUNTRIES, BASED ON MONTHLY IMF INTERNATIONAL FINANCIAL STATISTICS DATA ON OFFICIAL RESERVES EXCLUDING GOLD (DECEMBER 2024). THE VERTICAL LINE IS THE CAP UNDER A TOTAL AUM OF 100 BN. THE RIGHT PANEL ILLUSTRATES AN $\alpha = 0.5$ RULE UNDER WHICH EACH COUNTRY'S SUBSCRIPTION EQUALS THE MINIMUM BETWEEN 1.5 TIMES ITS RESERVES AND THE CAP. WHEN SUBSCRIPTIONS EXCEED INITIAL RESERVES, THE EXCESS IS INTERPRETED AS NEW SAVINGS. BLUE BARS DENOTE THE ORIGINAL SUBSCRIPTION, AND GREEN BARS INDICATE THE CHANGE. SOURCE: IMF STAFF CALCULATIONS.

$\eta = 0.5$ rule.[9] If the SRD lowers the cost of holding a liquid reserve claim, it can affect not only the composition of reserves but also their desired level.

7.3 *Market Depth, Composition, and Price Pressure*

To connect aggregate subscriptions to market impact, let w_k^* denote the weight on asset class k in the selected investment allocation from Section ??. The aggregate demand generated by subscriptions is then

$$D_k(\bar{S}, \eta) = w_k^* \mathscr{A}(\bar{S}, \eta). \tag{26}$$

For U.S. Treasuries, a simple reduced-form measure of market pressure compares this flow with either average daily trading volume ADV^{UST} or the outstanding stock O^{UST}:

$$\Pi_{UST}^{ADV} = \frac{D_{UST}}{ADV^{UST}}, \qquad \Pi_{UST}^{Stock} = \frac{D_{UST}}{O^{UST}}. \tag{27}$$

Using total AUM as a conservative upper bound for D_{UST}, the baseline US\$50 billion illustration amounts to less than 5 percent of one day's average U.S. Treasury trading volume and less

[9] A quadratic-preferences model in which the decision-maker values returns discounted for variance yields an allocation rule for the portfolio share proportional to $\frac{r}{\sigma^2\Gamma}$, where r is the expected return, σ is the standard deviation, and Γ a risk-aversion parameter. For the SRD returns, which would yield around an additional 2.59% annualized with a volatility of 3.36%, a value of $\eta = 0.5$ would correspond to a coefficient of risk aversion in the range $45 - 50$, which is high but reasonable for central bankers.

than 0.2 percent of the outstanding Treasury stock, given average daily trading volume above US$1 trillion and Treasury securities outstanding of about US$28.7 trillion in 2025Q2. At US$100 billion, the corresponding upper bounds remain below 10 percent of one day's turnover and about 0.35 percent of the outstanding stock. Because only a fraction of the pooled portfolio is allocated to Treasuries, the relevant Treasury-specific ratios are smaller still. But even assuming that the total AUM of US$50 (100) billion is effectively divested from Treasurys, this would represent a drop in demand for US debt equivalent to about 0.16% (0.33%) of GDP at end-2025 values. Plugging in the estimates from Krishnamurthy and Vissing-Jorgensen (2012) for the elasticity of Treasury yields implies an increase in yields of around 0.12 (0.25) basis points.[10] In addition, the price impact on short-term securities could be limited as these are implicitly anchored by monetary policy. All in all, under our baseline calibration, the implied direct pressure on U.S. sovereign yields is limited.

This does not imply that the SRD is irrelevant for asset demand. By lowering the quasi-fiscal cost of reserves, it can make reserve accumulation through a liquid pooled claim more attractive. The composition of demand would then shift away from a narrow concentration in short-dated bills toward a broader set of dollar-denominated assets, including longer-duration sovereign paper, supranational debt, and a tightly capped equity allocation. The relevant macroeconomic effect is therefore not a large displacement of Treasury demand, but an increase in the effective elasticity of the supply of instruments that satisfy reserve-use criteria. That is the channel through which the SRD can mitigate the pecuniary externalities associated with decentralized self-insurance (Caballero, Farhi, and Gourinchas, 2017).

Taken together, the aggregation exercises suggest that an SRD can matter for a sizable set of reserve holders even when it remains small relative to the depth of the core markets on which it relies. At such scales, the direct gains are primarily participant-level. Broader system-level effects would require either larger participation or a certification channel through which the vehicle makes diversified reserve-eligible claims more acceptable to decentralized reserve managers.

The SRD would complement, not replace, existing GFSN layers. Its direct function is to improve the micro and macro properties of the self-insurance layer of the GFSN. This could reduce the marginal need for precautionary holdings of the narrowest reserve assets and improve the efficiency of global liquidity provision. It would also create implementation dependencies, on legal recognition of the claim, governance of the vehicle, and reliability of any liquidity provider. These costs would need to be weighted against the benefits of the SRD. However, with estimated annual return improvements in excess of 200bps, there is ample room for these costs to be absorbed and yet generate a sizeable net gain.

[10] These result from a change in the residual debt/GDP ratio of 0.0017 (0.0033) log points.

8. Impact of the SRD on the Global Economy

If the SRD were large enough, it could affect global prices of assets and goods. Such spillovers could bolster or counter the beneficial effects of the fund described in the previous sections. In this section, we draw on ongoing work where we explore a theoretical framework for the global economy. The framework considers how and under which initial global asset configurations, the SRD can improve the GFSN without requiring Pigouvian taxes. In the model, the SRD can induce countries to shift their portfolio allocations and can also reduce transaction costs. The paper then investigates the general equilibrium effects of such a scheme.

8.1 Two Interacting Mechanisms

The framework is based on two mechanisms:

- Imperfect risk sharing by EMDE savers, which could reduce the interest rate on US short-term bonds if a large set of EMDEs suffer from these imperfections.
- A global financial cycle externality whereby the low interest rate on U.S. short-term bonds increases the global economy's vulnerability to U.S. financial shocks.

Imperfect risk sharing. In the model, EMDE savers (e.g., central banks) are assumed to be choosing the level of savings for their countries as a whole and allocating these savings across different asset classes. Each EMDE faces idiosyncratic income risk and their borrowing constraints become binding when their income is low. So, in those situations, they need to support consumption by drawing down their savings. They then prefer to save in safe short-term U.S. bonds whose value can be realized quickly with no transaction costs at moments of high urgency (following the criterion in Brunnermeier, Merkel, and Sannikov, 2024), instead of longer-dated assets (e.g., long-term U.S. bonds and U.S. or global equities) which may have a low value at those moments in addition to facing an array of transaction costs. This setup generates large redundancies in insurance, as each country over-provisions for their income shocks from a global point of view. EMDEs as a whole end up demanding a mix of securities which is more tilted toward short-term U.S. bonds than they would if their members could coordinate to insure each other against the income shocks.

This distortion to EMDEs' portfolios can significantly worsen their welfare and generate global spillovers. If global asset returns are such that returns to longer-dated assets exhibit a premium relative to those on short-term U.S. bonds, EMDE savers end up forgoing those returns as a result of imperfect insurance, and reserve accumulation is costly. Moreover, if a large set of

EMDEs suffer from this kind of imperfect risk sharing, the portfolio distortions generate additional global effects. The high demand for safe assets by EMDE savers then additionally pushes down the interest rate on U.S. short-term bonds.

The equilibrium distribution of global risks and the pricing of asset returns depends on which countries end up holding the risks. If the EMDEs prefer to hold less of the global risks, the U.S. must end up holding them and reaping the returns. In the model, this channel is manifested through the U.S. owning global financial intermediaries who borrow in U.S. short-term bonds to finance global investments, which is consistent with the global environment and mechanisms described by Gourinchas and Rey (2007) and Maggiori (2017).

Global financial cycle externality. The model features an endogenous aggregate risk mechanism that is built around the stylized choices of global firms. These firms form their initial financing decisions with banks based on the expected interest rates on loans. Accordingly, they disproportionately borrow from U.S. banks if they expect a low interest rate on U.S. short-term bonds. However, as their banking relationships are somewhat sticky, they are unable to quickly switch financial sources if the interest rates move later on, and they have to cut their investments in capital if those interest rates increase. As a result, when global firms are disproportionately borrowing from U.S. banks, shocks to these banks transmit disproportionately to asset prices and investment in the global economy. An important implication of this mechanism is that a more U.S.-centric set of financing sources ex-ante makes the global economy more risky to U.S.-bank-specific shocks ex-post, i.e., a U.S.-centered global financial cycle.

This externality is novel in terms of relating low U.S. interest rates to global aggregate risks. In our exploration of global agents' financing decisions, we draw on the insights of other frameworks without risk such as Jiang, Krishnamurthy, and Lustig (2024). In highlighting the problematic impact of low U.S. interest rates onto real variables, we agree with Das et al. (2022). That paper shows how EMDEs' FX mismatches are exacerbated. By contrast, we focus on spillovers to global firms as a whole, so that adverse spillovers are not confined primarily to EMDEs. As a result, a reduction of the externality in our framework is more likely to directly benefit not just EMDEs but also the U.S.

Interactions. Both mechanisms interact in the global general equilibrium. Imperfect risk sharing by a large set of EMDEs causes a large demand for U.S. short-term bonds and pushes down their interest rates, which in turn incentivizes global firms to disproportionately borrow from U.S. banks and causes the global economy to be highly sensitive to U.S.-bank-specific shocks. The amplified global financial cycle then further increases the volatility of longer-dated assets (e.g., long-term U.S. bonds and U.S. equities), making EMDEs shift even more away from these assets into U.S. short-term bonds as they desire more safety against global fluctuations.

8.2 *Pareto Improvements without Pigouvian Taxes*

Similar to Das et al. (2022), a global planner with an unrestricted set of tools would impose Pigouvian taxes on holdings of U.S. short-term bonds in order to redirect savings towards riskier longer-dated assets. This direct approach might in some cases decentralize the constrained-efficient allocation, but it is not realistic to imagine Pigouvian taxes being actually collected on EMDE central banks based on the composition and quantity of their reserve holdings.

The theoretical framework shows that under certain conditions, the SRD fund can take a more indirect route to generate Pareto improvements and dampen the global financial cycle by improving the set of assets available to EMDE central banks. Such benefits can arise if the fund can create a portfolio which satisfies the requirements of reserve assets, but that is not entirely made of assets that are themselves reserve assets (see Appendix F for more details). If so, it can offer higher returns at lower risk to the EMDE central banks, and thereby induce an incentive-compatible shift in these savers' portfolios away from U.S. short-term bonds to longer-dated securities. In the model thus far, we have considered the SRD fund investing in global equities.[11]

If the SRD fund is large, there can be additional global effects. As the SRD fund invests EMDEs' savings in global equities instead of U.S. short-term bonds, the U.S. short-term interest rate increases. As a result, the price of risk could change and global firms may also borrow less through U.S. banks. Both of these channels could weaken the U.S.-centered global financial cycle and thereby reduce the volatility of global capital, output, and equity prices. Any decrease in the amount of global risk could then enable EMDE savers to absorb even more equities and long-term bonds in their portfolios. However, the reduction in global risk does benefit firms, so aggregate gains can be large enough to overturn the price effects and generate a Pareto improvement.

Aggregate gains could be achieved by such a scheme without resorting to the extraordinary powers of Pigouvian taxes. However, the issue of distribution is not so trivial in the case of the SRD fund as it may be for tax-based interventions, since the proceeds from Pigouvian taxation could be used to compensate losers. In the case of the SRD fund, no revenues would be raised which could be used to share the aggregate gains. Rather, the theory suggests configurations of parameters under which the introduction of the SRD fund is likely to generate Pareto improvements. Moreover, currently the model features only global firms; separating them into U.S.- and EMDE-based firms could allow the SRD to invest disproportionately in U.S.-based firms as a way of generating a Pareto improvement (somewhat similarly to using the proceeds from Pigouvian taxation to redistribute the aggregate efficiency gains).

[11] If long-term U.S. bonds (or specifically dollar- or U.S. equities) were introduced into the model as another asset for the SRD fund to invest in, these bonds and equities would have the same returns properties as U.S. long-term bonds after U.S. interest rate shocks, so our results would remain robust.

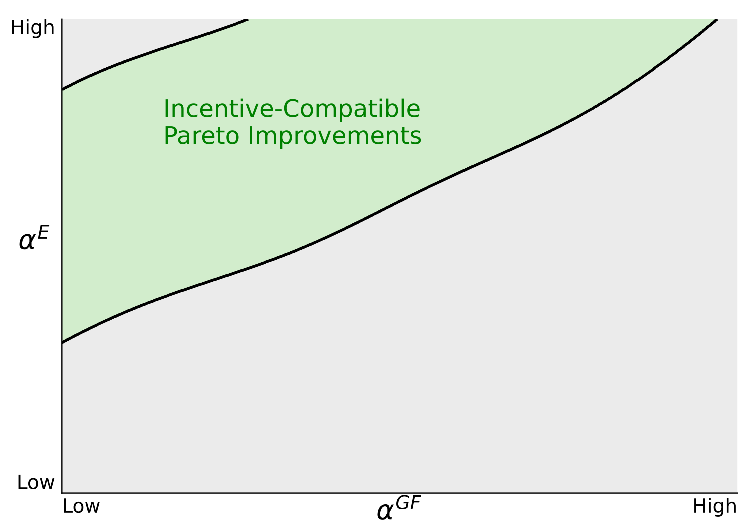
8.3 *Conditions Suggested by the Framework*

An SRD fund generates an *incentive-compatible Pareto improvement* if two criteria are satisfied: first, both EMDEs and the U.S. should benefit in net terms from an expected utility perspective; and second, EMDEs should prefer to save in the SRD fund instead of using their own portfolios which solely hold U.S. short-term bonds. The expected utility perspective means that all countries benefit on average across all possible future global outcomes; of course, with savings held in risky securities, the SRD portfolio would make losses after adverse shocks which reduce global equity prices, and those losses would be borne by the EMDE participants.

If the SRD fund is small and does not affect global asset prices and returns, it can generate an incentive-compatible Pareto improvement. The EMDE savers who participate in the fund gain from improved risk-sharing and higher returns, as there is a reduction in transaction costs. To keep an SRD fund small, it could be restricted to those EMDEs which have the lowest FX reserves and therefore the most severe asset diversification problems. It would not be targeted to EMDEs with large FX reserves, and this design could be justified because those EMDEs can already engage in diversification on their own, as they only need to draw down some portion of their reserves after stress events and can invest the rest in longer-dated securities. Given the small size of the fund, there would be no effect on the global financial cycle or U.S. welfare.

By contrast, if the SRD fund is large, it increases the U.S. short-term interest rate and may reduce global aggregate risks. Although these changes improve the chances of an overall increase in welfare, they also affect the likelihood of an incentive-compatible Pareto improvement.

FIGURE 11: INCENTIVE-COMPATIBLE PARETO IMPROVEMENTS.



Notes: α^E and α^{GF} are respectively the U.S.'s ownership shares of the global firms and financial intermediaries.
Source: IMF staff calculations.

The framework shows that an incentive-compatible Pareto improvement is more likely to occur if the global ownership of firms and financial intermediaries satisfies certain conditions. Those conditions are illustrated in Figure 11, a qualitative simulation which plots outcomes as a function of α^E and α^{GF}, respectively the U.S.'s ownership shares of the global firms and of the global financial intermediaries who borrow in U.S. short-term bonds to finance global investments.

- Incentive-compatible Pareto improvements exist in the green region where the U.S. owns enough of the global firms (i.e., α^E is high) relative to its holdings of global financial intermediaries (i.e., α^{GF} is low). In this case, EMDEs benefit from improved risk-sharing and lower global risks, while the U.S.'s benefits from those lower risks exceed its losses from the higher U.S. short-term interest rate.
- Some configurations of ownership are such that incentive-compatible Pareto improvements do not exist because either the U.S. or EMDEs end up worse off. If α^E is too low, meaning that the U.S. owns too few global firms, the U.S. does not benefit enough from the SRD fund's investments in global equities. If α^E is too high, the U.S. reaps too much of the gains from the SRD fund, leaving EMDEs worse off. If α^{GF} is too high, the U.S. owns too many of the global financial intermediaries and is too exposed to the short-term debt of those intermediaries, so it suffers disproportionately from the higher U.S. short-term interest rate that a large SRD fund generates.

Where the global economy fits into this picture depends on how we interpret the current configuration of global assets and liabilities and the set of assets that an SRD fund can invest in. The model is qualitative, and a more granular empirical implementation and calibration would be required to fully gauge which model configuration is most appropriate.

9. Conclusion

This paper studies whether part of reserve demand can be met through a pooled claim rather than through direct holdings of short-duration safe assets. The paper is based on the idea that portfolios which qualify as reserve assets need not be entirely made up of securities which are reserve assets. As a consequence, it is possible to find portfolios which offer higher returns than reserve assets but continue to satisfy the safety and liquidity properties of reserve assets. The paper proposes a Synthetic Reserve Deposit (SRD): a claim on a vehicle that combines a cash buffer with a diversified portfolio and, where needed, access to a collateralized liquidity window. The purpose is to relax the trade-off between liquidity and return without weakening reserve usability in periods of stress.

The paper makes three main points. First, the scheme with internalized investment management and contingent liquidity can improve the return profile of reserve holdings relative to a strategy concentrated in conventional reserve assets. Second, the illustrative SRD portfolio has return properties closer to those of reserve assets than to those of unconstrained risky assets. The SRD will make losses after jumps in interest rates on the bond categories and falls in equity prices, but the portfolio weights help reduce the size of such losses. Third, when liquidity is supplied against collateral rather than through forced sales, short-horizon tail risk can remain limited under conservative haircut and margining rules, so temporary valuation losses need not automatically disrupt liquidity access. Our illustrative exercises show that excess returns in the order of 250bps may be feasible under the assumed asset menu and downside-risk constraints. Under adequate design, the resulting claim can also satisfy reserve-asset recognition, ready-availability, legal, accounting, and operational requirements.

The general equilibrium impact of a large SRD is the topic of a companion paper. It explains the conditions under which an SRD can take an indirect route to generate Pareto improvements and dampen the global financial cycle by improving the set of assets available to EMDE central banks.

Several issues remain outside the scope of the paper. The quantitative exercises assume that the asset return distributions and correlations of different kinds of assets remain stable over time. They are also conducted in dollar assets and abstract from exchange-rate allocation, governance, legal form, accounting treatment, and the political economy of participation. These questions among others would need to be resolved if the illustrative exercise were to inspire practical implementation.

References

Acalin, J., M. Aguiar, S. Fourakis, C. Gutierrez, and A. Peralta-Alva (2026): "A tale of two defaults," Mimeo.

Benigno, G., L. Fornaro, and M. Wolf (2025): "The Global Financial Resource Curse," *American Economic Review*, 115, 220–62.

Brunnermeier, M. K., S. A. Merkel, and Y. Sannikov (2024): "Safe Assets," *Journal of Political Economy*, 132, 3603–3657.

Caballero, R. J., E. Farhi, and P.-O. Gourinchas (2017): "The Safe Assets Shortage Conundrum," *Journal of Economic Perspectives*, 31, 29–46.

Campbell, J. Y. (2000): "Asset Pricing at the Millennium," *The Journal of Finance*, 55, 1515–1567.

Cochrane, J. H. (2005): *Asset Pricing*, Princeton University Press.

Das, M., G. Gopinath, T. Kim, and J. C. Stein (2022): "Central Banks as Dollar Lenders of Last Resort: Implications for Regulation and Reserve Holdings," NBER Working Paper 30787, National Bureau of Economic Research.

Fornaro, L. and F. Romei (2019): "The Paradox of Global Thrift," *American Economic Review*, 109, 3745–79.

Gourinchas, P.-O. (2023): "International Macroeconomics: From the Great Financial Crisis to COVID-19, and Beyond," *IMF Economic Review*, 71, 1–34.

Gourinchas, P.-O. and M. Obstfeld (2012): "Stories of the Twentieth Century for the Twenty-First," *American Economic Journal: Macroeconomics*, 4, 226–265.

Gourinchas, P.-O. and H. Rey (2007): "From World Banker to World Venture Capitalist: US External Adjustment and the Exorbitant Privilege," in *G7 Current Account Imbalances: Sustainability and Adjustment*, ed. by R. H. Clarida, Univ. Chicago Press.

Gourinchas, P.-O. and H. Rey (2022): "Exorbitant Privilege and Exorbitant Duty," CEPR Discussion Papers 16944, C.E.P.R. Discussion Papers.

Hansen, L. P. (1982): "Large Sample Properties of Generalized Method of Moments Estimators," *Econometrica*, 50, 1029–1054.

Hansen, L. P. and S. F. Richard (1987): "The Role of Conditioning Information in Deducing Testable Restrictions Implied by Dynamic Asset Pricing Models," *Econometrica*, 55, 587–613.

International Monetary Fund (2013): "Revised Guidelines for Foreign Exchange Reserve Management," IMF Policy Paper, International Monetary Fund.

International Monetary Fund (2023): "The Global Financial Safety Net—A Stocktaking," IMF Policy Paper 2023/044, International Monetary Fund.

Jiang, Z., A. Krishnamurthy, and H. Lustig (2024): "Dollar Safety and the Global Financial Cycle," *The Review of Economic Studies*, 91, 2878–2915.

Krishnamurthy, A. and A. Vissing-Jorgensen (2012): "The Aggregate Demand for Treasury Debt," *Journal of Political Economy*, 120, 233–267.

Maggiori, M. (2017): "Financial Intermediation, International Risk Sharing, and Reserve Currencies," *American Economic Review*, 107, 3038–71.

Mendoza, E. G. and V. Quadrini (2024): "Macro-Financial Implications of the Surging Global Demand (and Supply) of International Reserves," Working Paper 32810, National Bureau of Economic Research.

World Bank (2023): "Reserve Management Survey Report 2023," Report, Reserve Advisory and Management Partnership (RAMP).

Appendices

A. Data Description and Asset Coverage

This appendix describes the asset universe used in the portfolio analysis. The asset menu combines benchmark fixed-income and equity return series. The fixed-income universe includes U.S. Treasury indices, USD-denominated sovereign and sub-sovereign bond indices, and USD-denominated investment-grade corporate bond indices. For sovereign and corporate bonds, the analysis includes A-, AA-, and AAA-rated indices with maturity buckets of 1–3 years, 1–5 years, and 1–10 years. For U.S. Treasuries, the corresponding 1–3-year, 1–5-year, and 1–10-year maturity segments are used. All bond return series are obtained from Haver Analytics through the `BONDINDX` database, which reports iBoxx USD indices. The equity return series correspond to the MSCI World and S&P 500 indices obtained from Bloomberg. Table 1 reports the full asset menu, together with the benchmark weights and the benchmark portfolio allocation.

The selected allocation combines four broad sources of exposure: U.S. Treasuries, USD-denominated sovereign and sub-sovereign bonds, USD-denominated investment-grade corporate bonds, and developed-market equities. As shown in Table 1, the largest allocation is to sovereign and sub-sovereign bonds, followed by corporate bonds, MSCI World equities, and U.S. Treasuries. Table 2 summarizes the geographic information available for each asset class. The Treasury sleeve represents direct U.S. government exposure. For the iBoxx fixed-income sleeves, geographic exposure should be interpreted as issuer-country or regional exposure rather than currency exposure, since all bond indices are denominated in U.S. dollars. Because public documentation does not provide country-level weights for each rating- and maturity-specific sub-index, the fixed-income geography is described using the broad iBoxx USD Benchmark universe. By contrast, the equity exposure is more directly identifiable through the MSCI World country composition.

Finally, Figure 12 reports the return correlation matrix across the assets included in the portfolio analysis. The figure shows that most fixed-income indices are highly correlated with one another, suggesting limited diversification within the bond universe. In contrast, equities display lower correlations with fixed-income assets, indicating that the equity sleeve provides the main source of cross-asset diversification in the benchmark portfolio.

Table 1: Assets Used in the Portfolio Analysis

Segment	Ticker	Benchmark (%)	Portfolio (%)
Treasuries			
1–3Y	IC766R@BONDINDX	33.3	10.00
1–5Y	IC873R@BONDINDX	33.3	0.00
1–10Y	IC659R@BONDINDX	33.3	0.46
Sovereigns and sub-sovereigns			
A 1–3Y	I7H13R@BONDINDX	–	0.00
A 1–5Y	I7J37R@BONDINDX	–	0.00
A 1–10Y	I7G06R@BONDINDX	–	49.00
AA 1–3Y	I7V56R@BONDINDX	–	0.06
AA 1–5Y	I7W63R@BONDINDX	–	0.28
AA 1–10Y	I7T35R@BONDINDX	–	0.00
AAA 1–3Y	I8676R@BONDINDX	–	0.46
AAA 1–5Y	I8783R@BONDINDX	–	1.17
AAA 1–10Y	I8569R@BONDINDX	–	0.23
Corporates			
A 1–3Y	IFW60R@BONDINDX	–	0.01
A 1–5Y	IFX77R@BONDINDX	–	0.13
A 1–10Y	IFV53R@BONDINDX	–	20.18
AA 1–3Y	IG787R@BONDINDX	–	0.76
AA 1–5Y	IG894R@BONDINDX	–	0.00
AA 1–10Y	IG670R@BONDINDX	–	0.42
AAA 1–3Y	IGL21R@BONDINDX	–	0.05
AAA 1–5Y	IGM38R@BONDINDX	–	0.00
AAA 1–10Y	IGK14R@BONDINDX	–	0.01
Equities			
MSCI World	MXWO Index	–	16.77
S&P 500	SPX Index	–	0.00

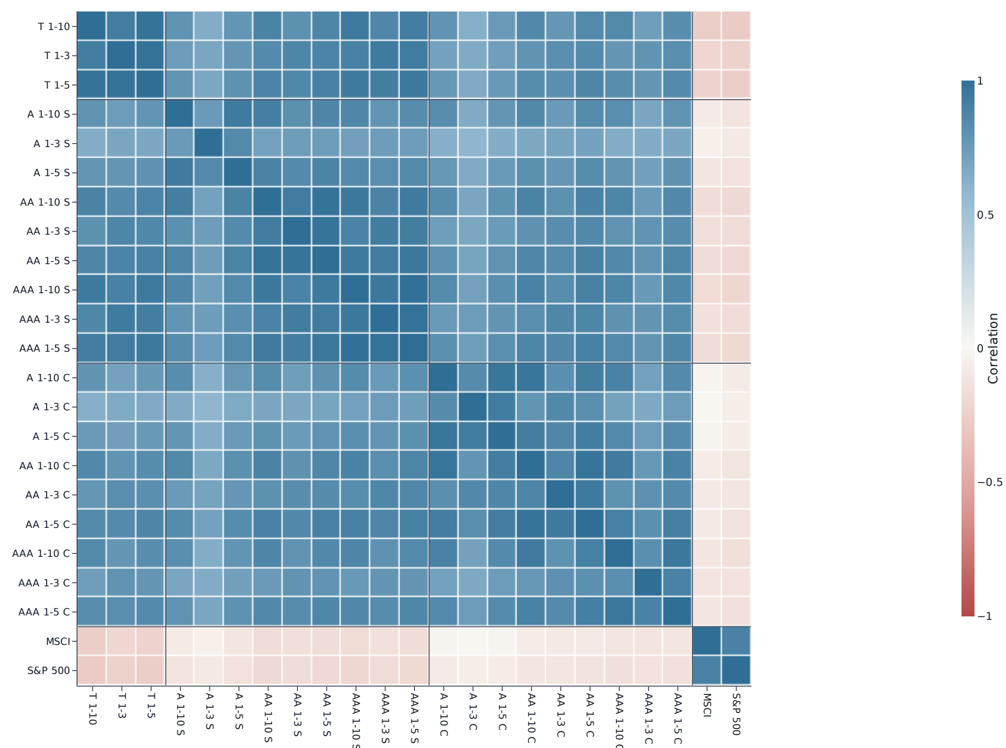
Notes: Treasury benchmark weights are equally distributed across the three maturity buckets. Bond indices are iBoxx USD series from Haver Analytics; equity indices are obtained from Bloomberg.

Table 2: Geographic Information Available for the Portfolio Asset Classes

Asset class / index family	Portfolio share (%)	Geographic exposure available from public documentation	Scope
U.S. Treasuries	10.46	United States. The Treasury sleeve consists of U.S. government securities in the 1–3Y and 1–10Y maturity buckets selected by the optimization.	Exact for this sleeve.
Sovereigns and sub-sovereigns	51.20	The relevant iBoxx USD family includes USD-denominated sovereign and sub-sovereign issuers, including sovereigns, agencies, local governments, supranationals, and other quasi-sovereign borrowers. Public iBoxx documentation reports the following regional composition for the broad iBoxx USD Benchmark universe: North America 85.4%, Asia 6.0%, Europe 5.4%, supranationals 1.8%, South America 0.7%, and Oceania 0.6%.	Approximate; broad iBoxx USD universe.
Corporate bonds	21.56	The corporate sleeve consists of USD-denominated investment-grade corporate issuers in the A, AA, and AAA rating buckets. Public documentation does not provide country weights for each corporate rating/maturity sub-index. The broad iBoxx USD Benchmark universe is dominated by North America, followed by Asia, Europe, supranationals, South America, and Oceania, using the regional shares reported above.	Approximate; broad iBoxx USD universe.
MSCI World	16.77	The MSCI World Index covers developed-market equities. As of March 31, 2026, the reported country weights were United States 71.27%, Japan 5.69%, United Kingdom 3.84%, Canada 3.58%, France 2.58%, and other developed markets 13.04%. The full developed-market country set includes Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, the United Kingdom, and the United States.	Exact factsheet countries; top-five weights reported separately.
S&P 500	0.00	United States. The index is included in the asset menu, but it receives zero weight in the benchmark portfolio.	No portfolio exposure.

Notes: Portfolio shares aggregate the individual weights reported in Table 1. Geographic exposure for fixed-income assets should be interpreted as issuer-country or regional exposure, not currency exposure, since all bond indices are denominated in U.S. dollars. For iBoxx fixed-income indices, public documentation reports regional composition for the broad iBoxx USD Benchmark universe, but not country-level weights for each rating- and maturity-specific sub-index. The iBoxx regional composition is therefore used only as an indicative description of the underlying USD bond universe. Exact country weights for the sovereign, sub-sovereign, and corporate sleeves would require constituent-level files for the relevant rebalance date. MSCI World country weights are reported directly in the MSCI factsheet as of March 31, 2026.

Figure 12: Return Correlation Matrix Across Assets



Notes: The figure reports pairwise correlations of asset returns across the full asset universe used in the portfolio analysis. Darker cells indicate stronger positive correlations.

B. Hybrid Semi-Parametric Simulation and Portfolio Selection

This appendix describes the simulation and portfolio-selection procedure that underlies the quantitative illustration in the main text. The objective is not to estimate a fully structural model of all asset returns. Rather, the aim is to construct a simulation device that is rich enough to evaluate medium-horizon portfolio outcomes for a reserve-style investment pool. The implementation is therefore best interpreted as a *hybrid semi-parametric Monte Carlo design.*

We proceed in five steps. First, daily return data are assembled and preprocessed. Second, each marginal distribution is estimated semi-parametrically and a multivariate Student-t copula is fitted in order to capture heavy-tailed cross-sectional dependence across the asset universe. Third, a separate vector autoregression is estimated for two selected benchmark bond return series in order to preserve explicit time-series dynamics for the benchmark block. Fourth, the simulated daily panel is aggregated to monthly returns and converted into rolling 36-month portfolio paths. Fifth, candidate portfolios are contrasted to a Treasury benchmark, and the fan chart in the main text is constructed from percentiles of those rolling paths.

Two features of the design deserve emphasis at the outset. A pure copula simulation would provide a flexible joint cross-section, but, absent an additional dynamic layer, it would treat daily observations as serially independent. Because the portfolio comparison in the paper is conducted over rolling multi-month horizons, persistence in benchmark fixed-income returns is economically relevant. Second, the rolling-window construction is also deliberate. The fan chart is not based on one arbitrarily chosen starting month. Instead, it is built from all admissible overlapping 36-month windows in the simulated monthly panel, which reduces the dependence of the results on a single initial date and yields a horizon distribution rather than a fixed-origin forecast.

B.1 Historical data and preprocessing

The input data consist of daily simple returns for a set of N assets observed between January 2003 and December 2024. Let r_{it}^{d} denote the daily return on asset i at date t. The historical sample is first restricted to that period, unavailable observations are treated as missing, and missing daily returns are linearly interpolated to obtain a balanced panel for the simulation exercise. The resulting daily return vector is

$$R_t^d = \left(r_{1t}^d, \dots, r_{Nt}^d\right)' . \tag{28}$$

This historical daily panel is used both to estimate the simulation model and, later, to compare historical and simulated moments.

B.2 Empirical marginal transformation and copula estimation

To model dependence flexibly while remaining agnostic about the exact parametric form of each marginal return distribution, each daily return series is first transformed into pseudo-uniform observations using its empirical cumulative distribution function. For asset i, let $\widehat{F}_i(\cdot)$ denote the empirical distribution of historical daily returns. The transformed observations are

$$U_{it} = \min\left\{\max\left[\widehat{F}_i\left(r_{it}^d\right), 10^{-6}\right], 1 - 10^{-6}\right\}. \tag{29}$$

The truncation away from zero and one avoids numerical difficulties in copula estimation.

Let $U_t = (U_{1t}, \dots, U_{Nt})'$. The cross-sectional dependence structure of U_t is then modeled with a multivariate Student-t copula,

$$U_t \sim \mathscr{C}(\cdot; \mathbf{R}, \nu), \tag{30}$$

where $\mathbf{R}$ is the copula correlation matrix and ν denotes the degrees of freedom. Both parameters are estimated by approximate maximum likelihood. This step allows the simulation to retain nonlinear dependence and tail comovement across the full set of assets without imposing Gaussianity.

B.3 VAR-based simulation for selected benchmark bond series

The copula is used to capture contemporaneous cross-sectional dependence, but the exercise also requires a simple device for preserving time-series persistence in the benchmark fixed-income block. For that reason, we estimate a separate vector autoregression for two selected benchmark bond return series, denoted b_{1t} and b_{2t}. These correspond to `IC766R@BONDINDX` and `MSCIR@BONDINDX`.

Stacking the two series into the vector

$$b_t = \begin{pmatrix} b_{1t} \\ b_{2t} \end{pmatrix}, \tag{31}$$

the procedure estimates

$$b_t = c + \sum_{\ell=1}^{252} A_\ell b_{t-\ell} + \varepsilon_t. \tag{32}$$

Thus, the MATLAB command `varm(2,252)` implies a two-equation VAR with 252 lags. In economic terms, this block is intended to preserve serial dependence in the benchmark fixed-income component over the horizons relevant for the portfolio comparison.

After estimation, the code simulates 10,000 daily observations from the fitted VAR. The script then applies an additional adverse-shock adjustment to a random subset of simulated observations. The resulting simulated returns for the two benchmark series are mapped into the unit interval using fitted Student-t location-scale marginals,

$$\widetilde{U}_{jt} = \widehat{G}_j\left(\tilde{b}_{jt}\right), \qquad j \in \{1, 2\}, \tag{33}$$

where $\widehat{G}_j(\cdot)$ denotes the fitted marginal cumulative distribution for benchmark series j.

B.4 Hybrid reconstruction of the full simulated return panel

The hybrid step combines the two ingredients above. Separately from the VAR simulation, the procedure draws 10,000 observations from the estimated multivariate Student-t copula for the full asset universe. Let U_t^c denote these copula draws. The first two copula margins are then replaced by the VAR-based uniforms $\left(\widetilde{U}_{1t}, \widetilde{U}_{2t}\right)$, so that the final simulated uniform vector is

$$U_t^* = \left(\widetilde{U}_{1t}, \widetilde{U}_{2t}, U_{3t}^c, \dots, U_{Nt}^c\right)'. \tag{34}$$

Each uniform draw is then transformed back into return space using the inverse of a fitted Student-t location-scale distribution for each asset,

$$\tilde{r}_{it}^d = \widehat{G}_i^{-1}\left(U_{it}^*\right), \qquad i = 1, \dots, N. \tag{35}$$

To align first moments, the implementation applies an asset-by-asset mean correction,

$$\tilde{r}_{it}^{d,\,\mathrm{adj}} = \tilde{r}_{it}^d + \left(\bar{r}_{i,\mathrm{hist}}^d - \bar{r}_{i,\mathrm{sim}}^d\right), \tag{36}$$

where $\bar{r}_{i,\mathrm{hist}}^d$ and $\bar{r}_{i,\mathrm{sim}}^d$ are the historical and simulated daily means for asset i, respectively.

This construction should be interpreted as a pragmatic compromise: the copula supplies a flexible heavy-tailed dependence structure for the broad asset universe, while the VAR imposes explicit time-series dynamics on the benchmark block that matters most for the downside comparison. But the procedure is not a single fully joint multivariate model.

A pure copula simulation would make the joint cross-section cleaner and easier to interpret, since all assets would then be generated from one common dependence structure. The trade-off is that, without an additional dynamic layer, daily returns would be serially independent and the rolling 36-month benchmark paths would not inherit their own time-series persistence. The present hybrid design accepts this trade-off in order to make the benchmark block more realistic at the horizon relevant for the paper.

B.5 Aggregation to monthly returns

Daily returns are converted into monthly returns by grouping observations into blocks of 21 trading days. For asset i in month m, monthly returns are computed as

$$R_{im} = \prod_{d=1}^{21} \left(1 + \tilde{r}_{i,m,d}^{d,\,\mathrm{adj}}\right) - 1. \tag{37}$$

The same aggregation is applied to the historical daily series in order to obtain a monthly benchmark for diagnostic comparisons.

B.6 Random portfolio generation

The procedure generates two classes of random portfolios. First, it constructs 10,000 candidate portfolios over the full asset universe. Let $w_p = (w_{p1}, \dots, w_{pN})'$ denote the weights of portfolio p. The first asset is assigned a fixed weight of 0.10, while the remaining 0.90 is allocated across assets 2 through N using normalized gamma draws. This is equivalent to a Dirichlet-type random weighting scheme with concentration parameter 0.1 and therefore tends to generate relatively concentrated portfolios.

Second, the procedure generates 500 benchmark portfolios using only the first three Treasury assets. These benchmark portfolios also place a fixed weight of 0.10 on the first asset and distribute the remaining 0.90 across assets 2 and 3. Monthly portfolio returns are then given by

$$R_{pm} = w_p' R_m, \tag{38}$$

where R_m is the vector of simulated monthly asset returns in month m.

B.7 Risk-return evaluation and portfolio selection

For each simulated portfolio, we compute annualized mean return and annualized volatility from the monthly return series:

$$\mu_p^{\mathrm{ann}} = (1 + \bar{R}_p)^{12} - 1, \qquad \sigma_p^{\mathrm{ann}} = \sigma(R_p)\sqrt{12}. \tag{39}$$

The central performance object in the paper, however, is based on rolling 36-month portfolio paths rather than on a single fixed starting date. Starting from an initial index level of 100, the procedure constructs

$$I_{p,s,1} = 100(1 + R_{p,s}), \tag{40}$$

and, for horizons $h = 2, \dots, 36$,

$$I_{p,s,h} = I_{p,s,h-1}(1 + R_{p,s+h-1}), \tag{41}$$

for all admissible starting dates s. The corresponding 36-month cumulative return is

$$C_{p,s}^{(36)} = \prod_{j=0}^{35}(1 + R_{p,s+j}) - 1, \tag{42}$$

and the associated annualized return is

$$A_{p,s}^{(36)} = \left(1 + C_{p,s}^{(36)}\right)^{12/36} - 1. \tag{43}$$

This rolling-window construction has two purposes. It transforms one long simulated monthly panel into many admissible 36-month investment histories, and it reduces sensitivity to an arbitrary start date. The resulting distribution is therefore best interpreted as the distribution of outcomes faced by an investor entering at an unspecified month, rather than as a single forecast path beginning from one fixed origin.

The implementation labels its downside metric as "VaR," but operationally it is the fifth percentile of the distribution of annualized 36-month returns,

$$\mathrm{VaR}_p = Q_{0.05}\left(A_{p,s}^{(36)}\right). \tag{44}$$

This object is thus a lower-tail return quantile rather than a positive loss measure, so larger values correspond to better downside performance.

The selected investment portfolio is obtained by comparing the 10,000 candidate portfolios against the set of Treasury benchmark portfolios. Candidate portfolios are retained only if three restrictions are satisfied: the combined weight on the two equity indices remains below 20 percent; the portfolio's fifth-percentile annualized rolling return exceeds the median counterpart among the Treasury benchmark portfolios; and the portfolio's mean annualized rolling return also exceeds the median among the Treasury benchmark portfolios. Among the admissible candidates, the selected investment allocation is the one with the highest mean annualized return.

B.8 The T-Bill Benchmark Portfolio

The T-bill benchmark used in Section 5 is built from the three short- to medium-duration U.S. Treasury indices that head the asset menu in Appendix A: the 1–3, 1–5, and 1–10 year Treasury return series.

We draw $K_B = 500$ long-only allocations on these three assets. In every draw, the 1–3 year Treasury receives a fixed weight of 0.10, and the remaining 0.90 is distributed across the 1–5 and

1–10 year indices using normalized $\Gamma(0.1, 1)$ draws. Daily returns for each index come from the same Student-t copula simulation used for the SRD portfolios (Appendix B), so the benchmark is exposed to the same shocks. Daily returns are compounded into monthly returns and aggregated at the portfolio level as

$$R_m^{B,(k)} = \sum_{i=1}^{3} w_i^{(k)} R_m^{(i)}. \tag{45}$$

For each draw we compute the mean and the fifth percentile of its annualized 36-month rolling returns. The representative benchmark $k^\star$ is the draw whose fifth percentile is closest to the cross-sectional median; its median return $\tilde{\mu}^B$ and median fifth percentile $\widetilde{\text{VaR}}^B$ define the admissibility bar for candidate SRD portfolios. The same $k^\star$ provides the comparison path in the fan chart of Figure 4 and the reference index for the time-to-outperform exercise of Figure 5.

B.9 Fan chart construction

The final fan chart compares the selected investment portfolio with the Treasury benchmark portfolio whose downside metric is closest to the median downside metric within the Treasury-only benchmark set. For both portfolios, the implementation computes pointwise percentiles of the rolling 36-month index paths and uses the 10th, 25th, 50th, 75th, and 90th percentiles to form the shaded bands and median lines shown in the main text.

C. Constructing the Stochastic Discount Factor

In asset pricing theory, the stochastic discount factor (SDF), also known as the pricing kernel, is a central object that links asset prices to their future payoffs. The fundamental pricing condition requires that for any asset i, the expectation of the product of the SDF and the asset's gross return equals one:

$$\mathbb{E}[M_{t+1} R_{t+1}^{(i)}] = 1.$$

Under the assumption of no arbitrage, this condition characterizes the pricing relation that the SDF must satisfy across assets. The SDF can be interpreted as a state-dependent weight that reflects the marginal value of payoffs in different states of the world. In economic terms, it corresponds to the intertemporal marginal rate of substitution of a representative investor (Cochrane, 2005; Campbell, 2000). Assets that deliver high payoffs in high-SDF (i.e., high marginal utility) states are more valuable, as they provide insurance against bad times. Conversely, assets that perform well in low-SDF states must offer higher average returns to compensate for their risk.

In the absence of exogenous factor data, we construct the SDF as a linear combination of returns on a small set of basis assets. This approach is standard in empirical asset pricing and is particularly useful in incomplete markets, where the full state space is not spanned by traded assets (Hansen and Richard, 1987). Specifically, we use the one-month returns on the 1–3 year U.S. Treasury index and the S&P 500 index as proxies for safe and risky factors, respectively. The SDF is specified as:

$$M_t = a + \gamma_1 R_t^{(1)} + \gamma_2 R_t^{(2)},$$

where a is a constant and γ_j are coefficients to be estimated. This specification treats the SDF as a linear factor model in returns. The choice of basis assets reflects the idea that a small number of traded portfolios can span the relevant sources of systematic risk (Cochrane, 2005).

To estimate the parameters (a, γ_1, γ_2), we impose the pricing condition on the universe of assets described in Appendix A and form moment conditions of the form:

$$\mathbb{E}[(M_t R_t^{(i)}) - 1] = 0.$$

These moment conditions are estimated using the Generalized Method of Moments (GMM), which minimizes the quadratic form:

$$J(\theta) = \bar{m}(\theta)^\top W \bar{m}(\theta),$$

where $\bar{m}(\theta)$ is the vector of sample moments and W is a weighting matrix (Hansen, 1982). GMM is well-suited for this setting, as it allows for overidentification (more moment conditions than parameters).

Table 3: GMM Estimates of the Stochastic Discount Factor

	Estimate	Std. Error	t-statistic	p-value
Constant (a)	1.05	3.66	28.5	0.00
Treasury factor (γ_1)	5.44	1.79	0.30	0.76
Equity factor (γ_2)	-5.37	1.37	-3.90	0.00

Notes: The stochastic discount factor is estimated as

$$M_t = a + \gamma_1 R_t^{\text{Treasury}} + \gamma_2 R_t^{\text{S\&P500}},$$

using iterative GMM applied to 23 asset-pricing moment conditions. Standard errors are HAC-robust. The Hansen overidentification test rejects the null that the model jointly satisfies the moment restrictions ($J = 288.47$, $p < 0.001$), so the specification should be interpreted as a reduced-form approximation. Coefficient magnitudes depend on the scaling of the basis-asset returns and are therefore not directly economically interpretable.

Table 3 shows that the estimated SDF loads positively and significantly on the Treasury factor, while the loading on the S&P 500 factor is negative but not statistically significant at conventional levels. Once estimated, the SDF $\hat{M}_t$ serves as an empirical pricing kernel. It can be used to assess the state-contingent behavior of asset returns. In particular, the covariance between an asset's return and the estimated SDF determines its exposure to aggregate risk. Assets with negative covariance with $\hat{M}_t$ tend to perform poorly in high marginal utility states and thus command higher expected returns. Assets with positive covariance provide insurance and earn lower premia.

D. Efficient Frontier

Figure 13 presents the opportunity set generated by the simulated portfolios using annualized returns and volatilities computed from monthly returns. The cloud of feasible portfolios displays the expected positive trade-off between risk and return, while the efficient frontier traces the upper envelope of attainable combinations. Portfolios with higher expected returns are generally associated with larger equity shares, as shown by the color gradient.

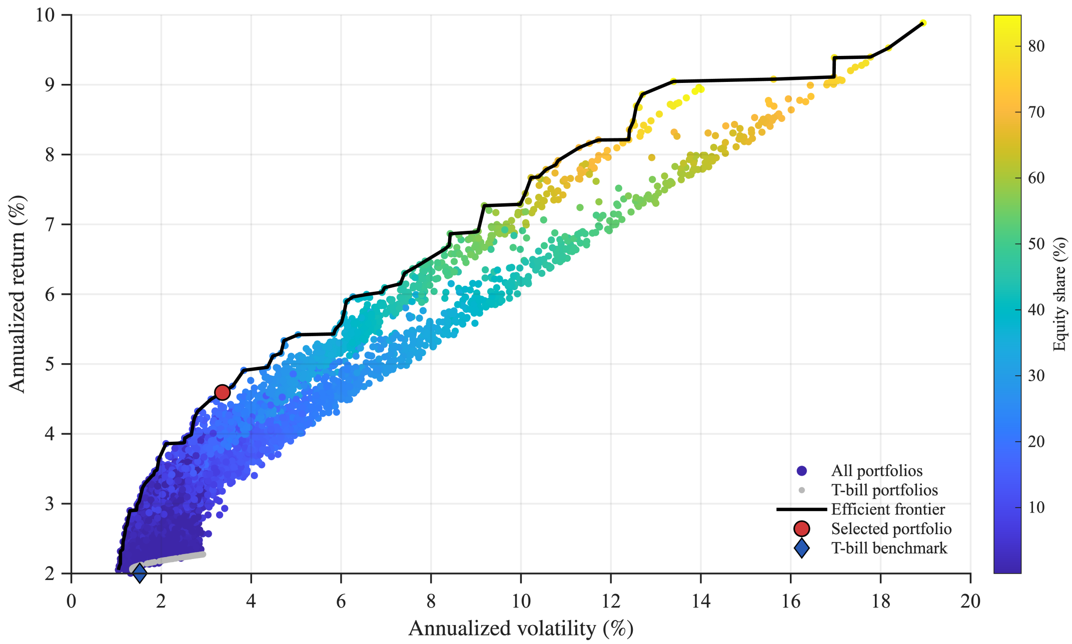
The benchmark portfolio lies close to the efficient frontier, indicating that it achieves a favorable risk-return trade-off relative to the broader feasible set. In particular, it delivers an annualized return of 4.59 percent with annualized volatility of 3.36 percent and a Sharpe ratio of 1.34. By contrast, the T-bill benchmark offers an annualized return of 2.00 percent, annualized volatility of 1.52 percent and a Sharpe ratio of 1.31. Importantly, the benchmark portfolio attains this improvement while maintaining limited equity exposure, at 16.8 percent of the portfolio.

E. Historical Counterfactual with Reserve Flows

The paper evaluates the asset side of the SRD by asking whether a diversified reserve-style portfolio can improve the return distribution relative to a Treasury benchmark. We now use the same portfolio to construct a liability-side backtest. The exercise asks what would have happened if an SRD had been introduced in 2007 and if participant countries had used their SRD accounts thereafter in the same way that they used reserves in the balance-of-payments data. The exercise is deliberately partial-equilibrium: it takes asset returns, reserve accumulation, and reserve drawdowns as observed, and asks whether the implied liquidity buffer would have been large enough to cover historical SRD use.

The counterfactual starts from the reserve stock in 2007. Countries deposit part of their foreign reserves into the SRD, and subsequent transactions follow quarterly balance-of-payments reserve-flow data. If a country's reserves increase in the data, the corresponding amount is treated as a deposit into the SRD. If reserves decline, the corresponding amount is treated as a withdrawal from the SRD account, subject to the account constraints described below. The sample excludes the United States, the euro area, and China, and is restricted to countries with quarterly

FIGURE 13: *EFFICIENT FRONTIER.*



Notes: The figure reports the opportunity set generated by simulated portfolios using annualized returns and volatilities computed from monthly returns. Each point corresponds to a feasible portfolio, with colors indicating the equity share. The solid line represents the efficient frontier. The benchmark portfolio and the T-bill benchmark are highlighted.

balance-of-payments data. This restriction follows the conservative sample used in the simulation and focuses the backtest on countries whose SRD withdrawals would be inferred from observed reserve-flow data.

Let A_t denote total SRD assets and let ℓ denote the liquidity-buffer ratio. In the counterfactual below we set

$$L_t = \ell A_t, \qquad \ell = 0.10, \tag{46}$$

so that 10 percent of SRD assets is held as the liquidity buffer. Total SRD assets are followed through time using the benchmark portfolio from Section ??, and 10 percent of the resulting asset value is treated as immediately liquid. We consider three subscription rules. The first is a full-transfer benchmark in which countries deposit all reserves into the SRD at inception. The second imposes a US$20 billion cap only on the initial deposit. Under this rule, the starting size of the SRD is limited, but subsequent reserve accumulation and portfolio returns can increase account values over time. The third imposes a US$20 billion cap on each SRD account at all dates. This rule limits concentration risk and total scale more tightly, but it also reduces the gains from pooling.

Figure 14 shows the implied liquidity buffer under the three rules. The full-transfer case produces the largest buffer by construction. The initial-cap case starts from a smaller pool but still grows substantially over time because the cap does not bind subsequent account balances. By contrast, the account-cap case keeps the buffer much smaller throughout the sample. The comparison therefore illustrates the main design trade-off. A looser cap preserves more of the pooling benefit, but allows the SRD to become large; a stricter account cap limits market footprint and concentration risk, but also limits the scale of pooled liquidity.

Figure 15 compares the buffer with realized withdrawals for the two capped designs. The green bars are aggregate withdrawals from SRD accounts implied by reserve drawdowns in the balance-of-payments data, while the lines report the corresponding 10 percent liquidity buffer. In both capped specifications, the buffer is large enough relative to observed SRD use over the sample. This includes the global financial crisis period, which generates a historically relevant stress episode for the backtest. The difference between the two panels is the extent of pooling. With only an initial cap, the buffer rises with account growth and remains far above observed withdrawals. With an account cap imposed at all dates, the buffer is much smaller and withdrawals are closer to the available liquidity, but the observed demands in the simulation remain covered.

The same logic can be pushed beyond observed reserve flows by comparing the buffer with simulated sudden-stop events. The stress exercise constructs historical sudden-stop episodes from quarterly gross-inflow data, expresses the resulting events in 2019 U.S. dollars, draws events without replacement, and compares the aggregate draw with the liquidity buffer implied by the

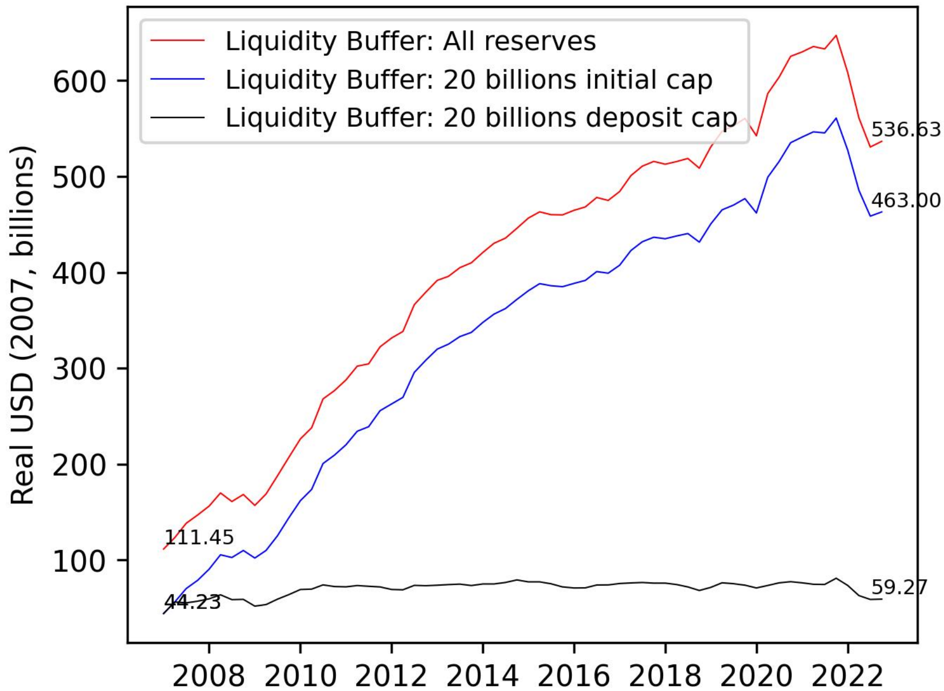


FIGURE 14: COUNTERFACTUAL LIQUIDITY BUFFER UNDER ALTERNATIVE SRD SIZE RULES.

Note: The counterfactual starts in 2007. Countries deposit reserves into the SRD and subsequent deposits and withdrawals follow quarterly balance-of-payments reserve-flow data. The liquidity buffer is 10 percent of SRD assets. The full-transfer case allows all reserves to be deposited. The initial-cap case limits only the initial deposit to US$20 billion per country. The account-cap case keeps each SRD account below US$20 billion at all dates. Amounts are in real 2007 U.S. dollars.

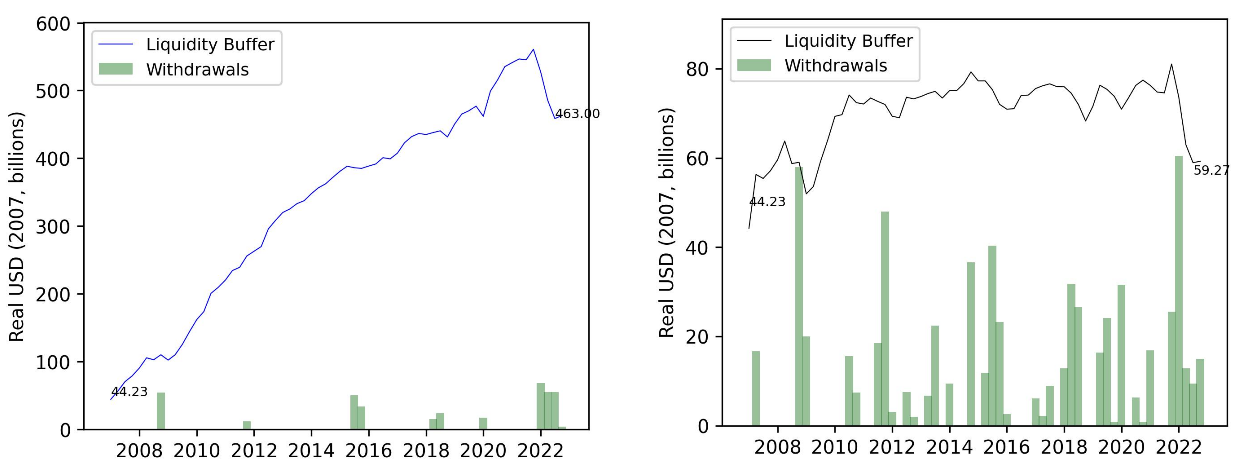


FIGURE 15: LIQUIDITY BUFFER AND WITHDRAWALS UNDER CAPPED SRD DESIGNS.

Note: The left panel applies a US$20 billion cap only to the initial deposit. The right panel applies a US$20 billion cap to SRD accounts at all dates. Green bars report withdrawals implied by reserve drawdowns in quarterly balance-of-payments data; solid lines report the 10 percent liquidity buffer under the corresponding rule. Amounts are in real 2007 U.S. dollars.

baseline SRD assets in 2019. Figure 16 varies the liquidity ratio on the horizontal axis and reports the share of sudden-stop draws covered by the resulting buffer. The exercise supports the view that a 10 percent buffer provides substantial coverage against GFC-sized sudden-stop shocks in the calibration, but not against arbitrarily larger shocks: coverage falls materially when the intensity of the shock is scaled up. The historical counterfactual should therefore be read as evidence that the 10 percent buffer is a reasonable calibration in the observed stress environment, not as a guarantee that the liquidity buffer would cover every possible systemic withdrawal scenario.

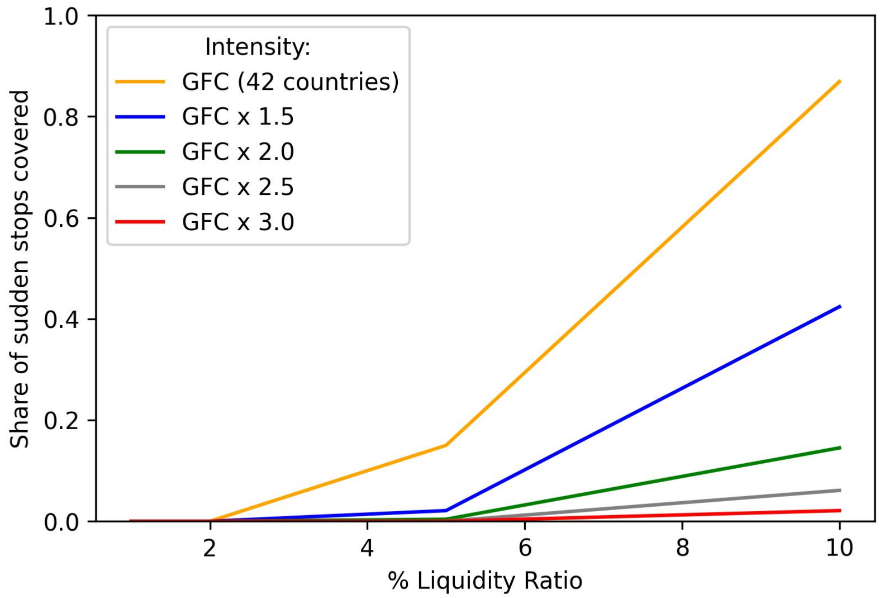


Figure 16: Sudden-stop coverage under alternative liquidity ratios.

Note: The liquidity buffer is constructed from SRD assets in 2019 under the baseline simulation. The horizontal axis varies the liquidity-buffer ratio. Sudden-stop events are constructed from quarterly gross-inflow data, converted to 2019 U.S. dollars, and drawn without replacement. Lines scale the intensity of the GFC-sized shock.

Taken together, the counterfactual supports two conclusions. First, once reserve liquidity is pooled, the first-tier buffer can be much larger than the realized use of SRD accounts even when the buffer is only 10 percent of assets. In the historical exercise, this is sufficient to avoid reliance on the contingent liquidity layer. Second, this result depends on the design of participation limits. An initial cap preserves the benefit of pooling but allows the SRD to grow, while a cap imposed at all dates contains size and concentration risk at the cost of a thinner buffer. The calibration of participation caps is therefore part of liquidity design, not only a market-impact constraint.

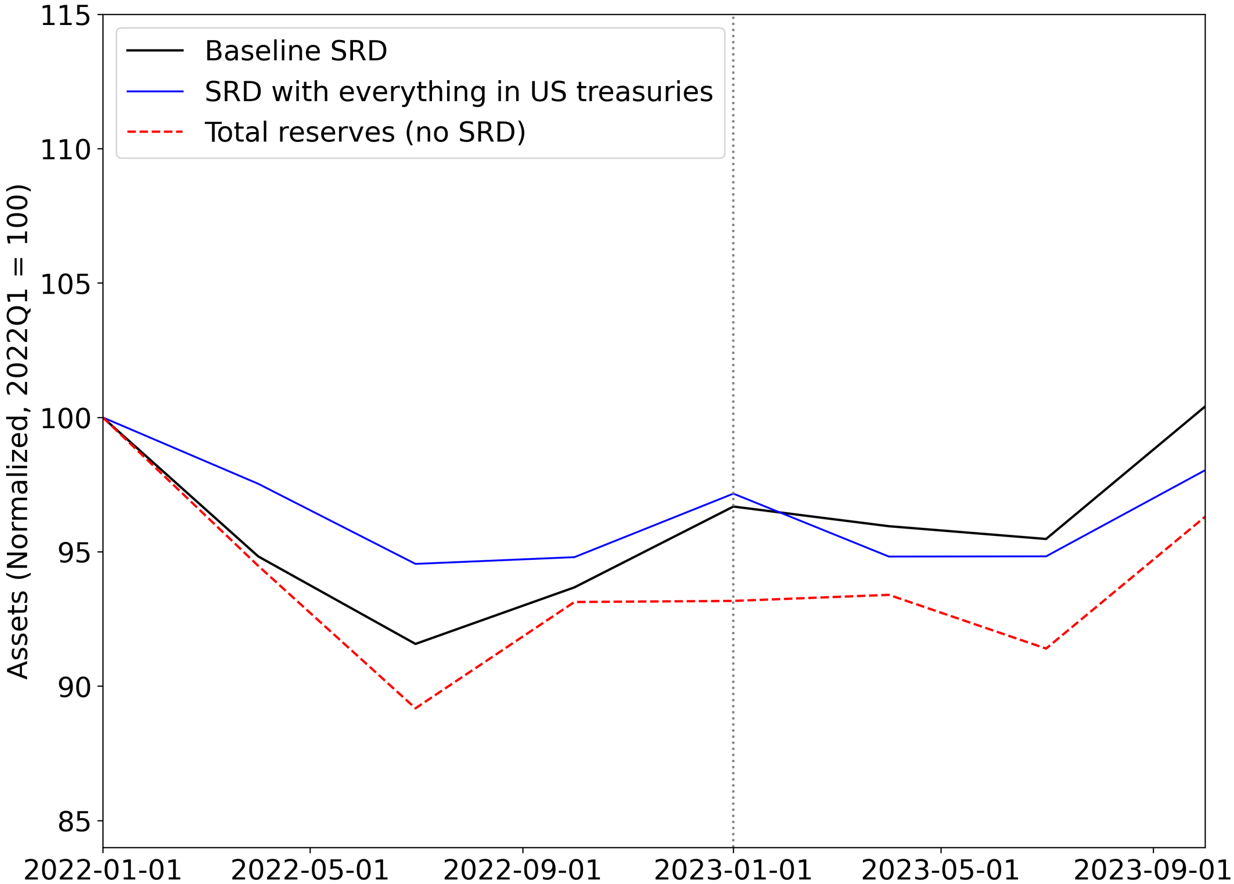


Figure 17: Performance of the benchmark portfolio during the SVB episode

F. Synthesizing Reserve Assets

Suppose there are two dates, $t = 0, 1$, and securities $j \in \{1, \ldots, N\}$, each with a distinct random return x_j at time $t = 1$. Suppose that reserve-asset status is determined by a function of these returns: asset j is a reserve asset if $\mathbb{E}[f(x_j)] \leq 0$, a natural example is that the function f measures how risky x_j is.[12] Assume that f is such that portfolios containing only reserve assets also satisfy the reserve-asset status. This means that $\mathbb{E}[f(\alpha x + (1-\alpha)v)] \leq 0$ when both $\mathbb{E}[f(x)], \mathbb{E}[f(v)] \leq 0$.

In this case, the portfolio choice problem of a central bank who must hold securities that are themselves reserve assets, and whose objectives are summarized by the function R, is

$$\begin{aligned} &\max_{\alpha_j \geq 0} \mathbb{E}\left[R\left(\sum_j \alpha_j x_j\right)\right] \\ \text{subject to} \quad &\sum_j \alpha_j = 1 \\ &\alpha_j = 0 \ \forall j : \mathbb{E}[f(x_j)] > 0 \end{aligned}$$

The requirement to only hold individual reserve assets can be understood by observing that it is much simpler to ensure that the portfolio is a reserve asset when it is entirely comprised of individual reserve assets; even if it is actively managed. On the other hand, if there were assets in the portfolio that are not reserve assets, continuous monitoring would be required to ensure that after each trade and valuation change, the portfolio shares continue to be such that the portfolio remains a reserve asset.

On the other hand, if the SRD scheme would choose the SRD portfolio and guarantee its reserve-asset status (for example by publicizing a set of constant portfolio shares which satisfy the reserve-asset definition), it would face a much more relaxed problem

$$\begin{aligned} &\max_{\alpha_j \geq 0} \mathbb{E}\left[R\left(\sum_j \alpha_j x_j\right)\right] \\ \text{subject to} \quad &\sum_j \alpha_j = 1 \\ &\mathbb{E}\left[f\left(\sum_j \alpha_j x_j\right)\right] \leq 0 \end{aligned}$$

This description clarifies two points. Firstly, to the extent that the function R puts some weight on returns, the SRD could achieve higher returns compared to a pure-reserve portfolio

[12] While we take an expectation, the function f is kept general to represent higher-order moments of the return; by the same token, we set a threshold of 0 which is fully arbitrary.

while retaining the safety and liquidity properties embedded in the function f. If R does not place a high weight on returns, but values something else entirely (e.g. it is even stricter than the reserve-asset definition in some particular way, for example with respect to downside risk), then the SRD could improve on those metrics. And second, once the SRD weights are publicized, individual central bank could hold the SRD portfolio at a much lower monitoring cost, which could also be a public good.